Group & Ring Theory MATP33: Course Summary of Important Results

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Lecture 0 — Background Results

Theorem 1 (First Isomorphism Theorem)**.** *Let $f : G \to H$ be a group homomorphism. Then* $\ker f \trianglelefteq G$ *and* $\operatorname{im} f \cong G/\ker f$*.*

Theorem 2 (Second Isomorphism Theorem)**.** *Let G be a group, $H \leq G$, and $N \trianglelefteq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.*

Theorem 3 (Third Isomorphism Theorem)**.** *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then $(G/K)/(L/K) \cong G/L$.*

Theorem 4 (Correspondence Theorem)**.** *Let $N \trianglelefteq G$. Every subgroup of G/N is of the form H/N where $N \leq H \leq G$, and conversely if $N \leq H \leq G$ then $H/N \leq G/N$.*

Theorem 5 (Lagrange)**.** *If $H \leq G$ and G is finite, then $|H|$ divides $|G|$.*

Theorem 6 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$ are non-zero non-units in $\mathbb{Z}$. This decomposition is unique.

Lecture 1

Theorem 7 (1.10)**.** *Let G be a group and X a set. Then:*

(i) If X is a G-set, then the action of G on X induces a homomorphism $\varphi : G \to S_X$;

(ii) Any homomorphism $\varphi : G \to S_X$ induces an action of G on X.

Proof. (i) Define $\varphi : G \to S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \to X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \operatorname{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G-set. □

Theorem 8 (1.11 — Cayley's Theorem)**.** *Every group G is isomorphic to a subgroup of the symmetric group S_G.*

Proof. By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \to S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \mathrm{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G/\ker \varphi \cong \operatorname{im} \varphi \leq S_G$. □

Definition 1 (Normal core)**.** *Let G be a group and $H \leq G$. The* normal core *of H in G is*

$$\mathrm{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

It is the largest normal subgroup of G contained in H.

Theorem 9 (1.12)**.** *Let G be a group and $H \leq G$ a subgroup of index n. Then there is a homomorphism $\varphi : G \to S_n$ such that $\ker \varphi = \mathrm{core}(H)$.*

Proof. Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of G on the coset space G/H by left multiplication gives a homomorphism $\varphi : G \to S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned}
\ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\
&= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\
&= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\
&= \bigcap_{x \in G} xHx^{-1} = \mathrm{core}(H).
\end{aligned}$$

□

Corollary 1 (1.13)**.** *Let G be a group with a normal subgroup H of index n. Then G/H is isomorphic to a subgroup of S_n.*

Proof. Since $H \trianglelefteq G$, conjugation fixes H, so $\mathrm{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \operatorname{im} \varphi \leq S_n$. □

Theorem 10 (1.24 — Orbit-Stabiliser Theorem)**.** *Let G be a group acting on a set X. Then the set of all orbits in X under G is a partition of X. Furthermore, for each $x \in X$ there is a bijection $Gx \to G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is finite,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where $C \subseteq X$ contains exactly one element from each orbit.

Proof. (Partition.) Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay$, $y = bz \Rightarrow x = (ab)z$. The equivalence class of x is exactly the orbit Gx, so the orbits partition X.

(Bijection $Gx \to G/G_x$.) Define

$$\varphi : Gx \to G/G_x, \qquad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset φ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence φ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. □

Theorem 11 (1.25 — Class Equation)**.** *Let G be a finite group. Then the conjugacy classes partition G, and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where $C \subseteq G$ contains exactly one element from each conjugacy class.

Proof. Let G act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of x under this action is its conjugacy class $\mathrm{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of x is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. □

Lecture 2

Theorem 12 (2.1)**.** *Let G be a finite group of order p^n, where p is prime and $n \in \mathbb{N}$. Then:*

(i) G has a non-trivial centre $Z(G)$;

(ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup K of G;

(iii) if H is a proper subgroup of G, then $H \subsetneq N_G(H)$. In particular, if $|H| = p^{n-1}$ then $H \trianglelefteq G$.

Proof. (i) Since $x \in Z(G)$ iff $\mathrm{ccl}(x) = \{x\}$, the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where C runs over conjugacy classes with more than one element. For each $x \in C$, $|C_G(x)|$ properly divides p^n, so $p \mid |G : C_G(x)|$. Hence p divides the sum, and therefore $p \mid |Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq 1$, so in fact $|Z(G)| \geq p > 1$.

(ii) Intersecting the class decomposition of G with K gives $|K| = |Z(G) \cap K| + \sum_{x \in C} |\operatorname{ccl}(x) \cap K|$. Since $K \trianglelefteq G$, each conjugacy class is either disjoint from K or contained in K, so $p \mid \sum |\operatorname{ccl}(x) \cap K|$. As $p \mid |K|$, we get $p \mid |Z(G) \cap K|$, and the same argument as (i) gives $Z(G) \cap K$ non-trivial.

(iii) Let M be a maximal normal subgroup of G contained in H. Then G/M is a p-group, so by (i) it has a non-trivial centre $L/M \trianglelefteq G/M$. By the Correspondence Theorem, $L \trianglelefteq G$. Since $M \subsetneq L$ strictly, maximality of M forces $L \not\subseteq H$.

It remains to show $L \subseteq N_G(H)$. To show $\ell \in N_G(H)$ we need $\ell^{-1}h\ell \in H$ for all $h \in H$. Take $h \in H$ and $\ell \in L$. Since $\ell M \in L/M = Z(G/M)$, it commutes with every coset, so $(\ell M)(hM) = (hM)(\ell M)$, i.e. $\ell hM = h\ell M$. This gives $\ell^{-1}h\ell \in hM$. Since $h \in H$ and $M \subseteq H$, we have $hM \subseteq H$, so $\ell^{-1}h\ell \in H$. Hence $\ell \in N_G(H)$, so $L \subseteq N_G(H)$.

Since $L \subseteq N_G(H)$ but $L \not\subseteq H$, we have $H \subsetneq N_G(H)$. Finally, if $|H| = p^{n-1}$ then $N_G(H)$ has order strictly greater than p^{n-1}, and the only possibility is $p^n = |G|$, so $N_G(H) = G$ and $H \trianglelefteq G$. □

Theorem 13 (2.2 — Burnside's Lemma)**.** *Let G be a finite group acting on a finite set X, and for $g \in G$ let $X_g := \{x \in X \mid gx = x\}$. Then the number k of orbits in X under G is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Proof. Let $S := \{(g, x) \in G \times X \mid gx = x\}$. Fixing g gives $|S| = \sum_{g \in G} |X_g|$; fixing x gives $|S| = \sum_{x \in X} |G_x|$. Let $C \subseteq X$ contain one element from each orbit. By the Orbit-Stabiliser Theorem, $|G_x| = |G|/|Gx| = |G|/|Ga|$ for all $x \in Ga$. Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \qquad \square$$

Remark 1 (Pólya counting)**.** *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with p beads (p prime) can be made from n colours, where two necklaces are the same if one is a rotation of the other?*

Let X be the set of all colourings, so $|X| = n^p$. The cyclic group $G = \langle a \rangle$ of order p acts on X by rotation. For the identity, $X_e = X$ so $|X_e| = n^p$. For any $g \neq e$: since $|G|$ is prime, g generates G, so a necklace fixed by g must be fixed by all rotations, meaning all beads have the same colour. Hence $|X_g| = n$. Burnside then gives

$$k = \frac{1}{p}\left(n^p + (p-1)n\right).$$

Lecture 3

Definition 2 (p-group, Sylow p-subgroup)**.** *Let p be a prime. A group G is a* p-group *if every element of G has order a power of p. If G is finite and $p^m \mid |G|$ but $p^{m+1} \nmid |G|$, then any subgroup of G of order p^m is called a* Sylow p-subgroup *of G.*

Theorem 14 (3.5 — First Sylow Theorem)**.** *Let G be a finite group and p a prime. If $p^m \mid |G|$, then G has a subgroup of order p^m.*

Proof. If $m = 0$ the result is trivial. We proceed by induction on $n = |G|$, the case $n = 1$ being trivial. Assume the result holds for all groups of order less than n.

Case 1: $p \mid |Z(G)|$. By the Fundamental Theorem of Finitely Generated Abelian Groups, $Z(G)$ contains an element a of order p. Let $C = \langle a \rangle$. Since $a \in Z(G)$, we have $C \trianglelefteq G$, and $|G/C| = n/p$, which is divisible by p^{m-1}. By the induction hypothesis, G/C has a subgroup of order p^{m-1}, which by the Correspondence Theorem is of the form H/C for some $H \leq G$. Then $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$.

Case 2: $p \nmid |Z(G)|$. From the class equation $n = |Z(G)| + \sum_a |G : C_G(a)|$, since $p \mid n$ but $p \nmid |Z(G)|$, there must exist some $a \notin Z(G)$ with $p \nmid |G : C_G(a)|$. Since $|G| = |G : C_G(a)| \cdot |C_G(a)|$ and $p^m \mid |G|$ but $p \nmid |G : C_G(a)|$, it follows that $p^m \mid |C_G(a)|$. Since $a \notin Z(G)$, we have $C_G(a) \neq G$, so $|C_G(a)| < n$. By the induction hypothesis $C_G(a)$ contains a subgroup of order p^m, which is also a subgroup of G. □

Corollary 2 (3.6 — Cauchy's Theorem)**.** *If p is a prime dividing $|G|$, then G has an element of order p.*

Proof. By the First Sylow Theorem, G has a subgroup H of order p. Every non-identity element of H has order p. □

Corollary 3 (3.7)**.** *A finite group G is a p-group if and only if $|G|$ is a power of p.*

Proof. If $|G|$ is a power of p, then by Lagrange every element order divides $|G|$, hence is a power of p. Conversely, if $|G|$ is not a power of p, then some prime $q \neq p$ divides $|G|$, and by Cauchy's Theorem G has an element of order q, contradicting that G is a p-group. □

Theorem 15 (3.8 — Second Sylow Theorem)**.** *Let G be a finite group and p a prime. All Sylow p-subgroups of G are conjugate.*

Proof. Let $S, H \in \mathrm{Syl}_p(G)$. Let S act on the coset space G/H by left translation: $s \cdot gH = sgH$.

Orbits have p-power size. By the Orbit-Stabiliser Theorem, each orbit has size $|S : S_{gH}|$, which divides $|S| = p^m$, hence is a power of p.

There must be a fixed point. Since H is a Sylow p-subgroup, $p \nmid |G|/|H| = |G/H|$. Writing $|G/H|$ as a sum of orbit sizes, each of which is a power of p, the only way this sum is not divisible by p is if at least one orbit has size $p^0 = 1$.

A fixed point gives conjugacy. If gH is a fixed point, then $s \cdot gH = gH$ for all $s \in S$, so $g^{-1}sg \in H$ for all $s \in S$. Hence $g^{-1}Sg \subseteq H$, and since $|g^{-1}Sg| = |S| = |H|$, we get $g^{-1}Sg = H$. So S and H are conjugate. □

Corollary 4 (3.10)**.** *A Sylow p-subgroup of G is unique if and only if it is normal in G.*

Proof. If P is the unique Sylow p-subgroup, then $gPg^{-1} = P$ for all $g \in G$ (since conjugates are also Sylow p-subgroups), so $P \trianglelefteq G$. Conversely, if $P \trianglelefteq G$ then P is its own conjugate, so by the Second Sylow Theorem it is the only Sylow p-subgroup. □

Remark 2. *The direction "unique ⇒ normal" does* not *use the Second Sylow Theorem (which only gives* some *g with $gPg^{-1} = Q$). Instead, for* every *$g \in G$ the map $x \mapsto gxg^{-1}$ is an injective homomorphism, so gPg^{-1} is a subgroup with $|gPg^{-1}| = |P|$, i.e. again a Sylow p-subgroup; uniqueness then forces $gPg^{-1} = P$. The Second Sylow Theorem is only needed for "normal ⇒ unique": any Sylow Q equals gPg^{-1} for some g, and normality gives $gPg^{-1} = P$.*

Theorem 16 (3.9 — Third Sylow Theorem)**.** *Let G be a finite group and p a prime. The number n_p of Sylow p-subgroups satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. $n_p \mid |G|$. By the Second Sylow Theorem, all Sylow p-subgroups are conjugate to a fixed $K \in \mathrm{Syl}_p(G)$, so $n_p = |\mathrm{ccl}(K)| = |G : N_G(K)|$, which divides $|G|$.

$n_p \equiv 1 \pmod{p}$. Let $P \in \mathrm{Syl}_p(G)$. Since conjugates of Sylow p-subgroups are again Sylow p-subgroups (by the Second Sylow Theorem), P acts on the set $\mathrm{Syl}_p(G)$ by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P,\ Q \in \mathrm{Syl}_p(G).$$

The orbit of Q is $P \cdot Q = \{xQx^{-1} : x \in P\}$, and its stabiliser is $\{x \in P : xQx^{-1} = Q\} = P \cap N_G(Q) = N_P(Q)$. By the Orbit-Stabiliser Theorem,

$$|P \cdot Q| = |P : N_P(Q)|,$$

which divides $|P| = p^m$ and is therefore a power of p. In particular $|P \cdot Q| = 1$ if and only if $N_P(Q) = P$.

Let $F = \{Q \in \mathrm{Syl}_p(G) : N_P(Q) = P\}$ be the set of fixed points, and choose representatives $Q_1, \ldots, Q_r$, one from each orbit of size greater than 1. The orbits partition $\mathrm{Syl}_p(G)$, so

$$n_p = |\mathrm{Syl}_p(G)| = |F| + \sum_{i=1}^{r} |P : N_P(Q_i)|.$$

Each $|P : N_P(Q_i)|$ is a power p^{e_i} with $e_i \geq 1$, hence divisible by p. Therefore $n_p \equiv |F| \pmod{p}$.

It remains to show $F = \{P\}$, which gives $n_p \equiv 1 \pmod{p}$. Clearly $P \in F$. If $Q \in F$, then $xQx^{-1} = Q$ for all $x \in P$, so $P \leq N_G(Q)$. By Corollary 3.10, since $Q \trianglelefteq N_G(Q)$, Q is the unique Sylow p-subgroup of $N_G(Q)$. But P is also a Sylow p-subgroup of $N_G(Q)$, so $P = Q$. Hence $F = \{P\}$. □

Lecture 4 — Modules: the Basics

Definition 3 (4.1 — Left R-module)**.** *Let R be a ring. An abelian group M is a* left R-module *if there is a map $R \times M \to M$, $(r, m) \mapsto rm$, such that for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$:*

(i) $r(m_1 + m_2) = rm_1 + rm_2$;

(ii) $(r_1 + r_2)m = r_1m + r_2m$;

(iii) $(r_1r_2)m = r_1(r_2m)$;

(iv) $1m = m$ *if* $1 \in R$.

Lemma 1 (4.7)**.** *Let M be an R-module. Then for $a \in R$ and $m \in M$:*

(i) $0m = 0$;

(ii) $a0 = 0$;

(iii) $(-a)m = -(am) = a(-m)$.

Proof. (i) $0m = (0+0)m = 0m + 0m$. Subtracting $0m$ from both sides gives $0 = 0m$.

(ii) $a0 = a(0+0) = a0 + a0$. Subtracting $a0$ from both sides gives $0 = a0$.

(iii) By (i) and axiom (ii): $0 = 0m = (a+(-a))m = am+(-a)m$, so $(-a)m = -(am)$. By (ii) and axiom (i): $0 = a0 = a(m + (-m)) = am + a(-m)$, so $a(-m) = -(am)$. □

Definition 4 (4.8 — Submodule)**.** *A subset N of an R-module M is an* R-submodule *of M if:*

(i) $N \neq \emptyset$*;*

(ii) $a - b \in N$ *for all* $a, b \in N$*;*

(iii) $ra \in N$ *for all* $a \in N$*,* $r \in R$*.*

Theorem 17 (4.12)**.** *Let $(N_i)_{i\in I}$ be a family of R-submodules of an R-module M. Then $\bigcap_{i\in I} N_i$ is also an R-submodule of M.*

Proof. Since $0 \in N_i$ for all i, we have $0 \in \bigcap N_i$, so it is non-empty. Let $x, y \in \bigcap N_i$ and $r \in R$. For each i, since N_i is a submodule, $x - y \in N_i$ and $rx \in N_i$. Hence $x - y$ and rx lie in $\bigcap N_i$. □

Definition 5 (4.13 — Generated submodule, cyclic module)**.** *Let $S \subseteq M$. The family $\mathcal{A} = \{N \mid N$ is an R-submodule of M containing $S\}$ is non-empty (since $M \in \mathcal{A}$), so by Theorem 4.12 the intersection $\bigcap_{N\in\mathcal{A}} N$ is an R-submodule. It is the* smallest *submodule of M containing S, called the submodule* generated by *S and denoted (S). If $S = \{x_1, \ldots, x_n\}$ we write $(x_1, \ldots, x_n)$. An R-module M is* cyclic *if $M = (x)$ for some $x \in M$.*

Theorem 18 (4.15)**.** *Let R be a ring with unity and $x_1, \ldots, x_n \in M$. Then*

$$(x_1, \ldots, x_n) = \sum_{i=1}^{n} Rx_i := \{r_1x_1 + \cdots + r_nx_n \mid r_i \in R\}.$$

Proof. ($\supseteq$) The submodule $(x_1, \ldots, x_n)$ contains each x_i. By closure under scalar multiplication it contains rx_i for all $r \in R$, and by closure under addition it contains every sum $r_1x_1 + \cdots + r_nx_n$. Hence $\sum Rx_i \subseteq (x_1, \ldots, x_n)$.

($\subseteq$) We show $\sum Rx_i$ is a submodule containing each x_i; then $(x_1, \ldots, x_n) \subseteq \sum Rx_i$ follows from minimality of $(x_1, \ldots, x_n)$.

Each $x_i \in \sum Rx_i$ since $x_i = 1 \cdot x_i$ (using unity). For the submodule conditions: $0 = \sum 0 \cdot x_i \in \sum Rx_i$, so it is non-empty. For $\sum r_ix_i, \sum s_ix_i \in \sum Rx_i$ and $r \in R$,

$$\sum r_ix_i - \sum s_ix_i = \sum (r_i - s_i)x_i \in \sum Rx_i, \qquad r\sum r_ix_i = \sum (rr_i)x_i \in \sum Rx_i. \quad \square$$

Definition 6 (4.18 — Sum of submodules)**.** *Let $N_1, \ldots, N_k$ be R-submodules of M. Their* sum *$\sum_{i=1}^{k} N_i$ is the submodule of M generated by $\bigcup_{i=1}^{k} N_i$, i.e. the smallest submodule of M containing each N_i.*

Lecture 5 — Modules: continuation

Theorem 19 (4.19)**.** *Let $N_1, \ldots, N_k$ be R-submodules of an R-module M. Then*

$$\sum_{i=1}^{k} N_i = \{x_1 + \cdots + x_k \mid x_i \in N_i\}.$$

Proof. Let

$$A = \{x_1 + \cdots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that A is a submodule of M, and it contains every N_i. Hence, by minimality,

$$\sum_{i=1}^{k} N_i \subseteq A.$$

Conversely, $\sum_{i=1}^{k} N_i$ contains every N_i and is closed under addition, so it contains every element of A. Therefore

$$A \subseteq \sum_{i=1}^{k} N_i.$$

□

Definition 7 (4.21 — Direct sum)**.** *The sum $N_1 + \cdots + N_k$ is* direct *if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \qquad x_i \in N_i.$$

In this case we write

$$N_1 \oplus \cdots \oplus N_k.$$

Theorem 20 (4.22)**.** *Let $N_1, \ldots, N_k$ be submodules of an R-module M. The following are equivalent:*

(i) $\sum_{i=1}^{k} N_i$ *is direct;*

(ii) if $\sum_{i=1}^{k} x_i = 0$*, where* $x_i \in N_i$*, then every* $x_i = 0$*;*

(iii) for every i*,*

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Proof. (i) ⇒ (ii): Zero already has the representation $0 + \cdots + 0$, so uniqueness implies that every $x_i = 0$.

(ii) ⇒ (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then $x = \sum_{j \neq i} x_j$ for some $x_j \in N_j$. Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives $x = 0$.

(iii) $\Rightarrow$ (i): Suppose

$$\sum_{i=1}^{k} x_i = \sum_{i=1}^{k} y_i, \qquad x_i, y_i \in N_i.$$

For every i,

$$x_i - y_i = -\sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus $x_i = y_i$ for every i, so the representation is unique.

□

Definition 8 (4.23 — R-homomorphism)**.** *Let M and N be R-modules. A map $f\colon M \to N$ is an* R-homomorphism *if, for all $x, y \in M$ and $r \in R$,*

$$f(x + y) = f(x) + f(y), \qquad f(rx) = rf(x).$$

The set of all R-homomorphisms $M \to N$ is denoted

$$\mathrm{Hom}_R(M, N).$$

An R-homomorphism $M \to M$ is called an endomorphism*, and*

$$\mathrm{End}_R(M) := \mathrm{Hom}_R(M, M).$$

Lemma 2 (Properties of homomorphisms)**.** *Let $f\colon M \to N$ be an R-homomorphism. Then for all $a, b \in M$ and $r \in R$:*

(i) $f(0_M) = 0_N$*;*

(ii) $f(a - b) = f(a) - f(b)$*;*

(iii) $r \cdot 0_N = 0_N$*.*

Proof. (i) By additivity, $f(0_M) = f(0_M + 0_M) = f(0_M) + f(0_M)$. Adding $-f(0_M)$ to both sides gives $0_N = f(0_M)$.

(ii) By additivity, $f(a - b) + f(b) = f\big((a - b) + b\big) = f(a)$. Adding $-f(b)$ to both sides gives $f(a - b) = f(a) - f(b)$.

(iii) By the module axiom $r(x+y) = rx+ry$, we have $r\cdot 0_N = r(0_N+0_N) = r\cdot 0_N+r\cdot 0_N$. Adding $-(r \cdot 0_N)$ to both sides gives $0_N = r \cdot 0_N$. □

Lemma 3 (Kernel and image are submodules)**.** *Let $f\colon M \to N$ be an R-homomorphism. Then*

$$\ker f = \{x \in M \mid f(x) = 0_N\} \quad \textit{and} \quad \mathrm{im}\, f = f(M) = \{f(x) \mid x \in M\}$$

are R-submodules of M and N respectively.

Proof. We check the three conditions of Definition 4.8, using parts (i)–(iii) of Lemma 2.

$\ker f \leq M$.

(i) By (i), $f(0_M) = 0_N$, so $0_M \in \ker f$ and $\ker f \neq \emptyset$.

(ii) Let $a, b \in \ker f$. By (ii), $f(a - b) = f(a) - f(b) = 0_N - 0_N = 0_N$, so $a - b \in \ker f$.

(iii) Let $a \in \ker f$ and $r \in R$. Since f is an R-homomorphism and by (iii), $f(ra) = rf(a) = r \cdot 0_N = 0_N$, so $ra \in \ker f$.

$\operatorname{im} f \leq N$.

(i) By (i), $0_N = f(0_M) \in \operatorname{im} f$, so $\operatorname{im} f \neq \emptyset$.

(ii) Let $u, v \in \operatorname{im} f$, say $u = f(a)$ and $v = f(b)$ with $a, b \in M$. By (ii), $u - v = f(a) - f(b) = f(a - b)$, and $a - b \in M$, so $u - v \in \operatorname{im} f$.

(iii) Let $u = f(a) \in \operatorname{im} f$ and $r \in R$. Since f is an R-homomorphism, $ru = rf(a) = f(ra)$, and $ra \in M$, so $ru \in \operatorname{im} f$. □

Theorem 21 (4.30)**.** *Let $M = \bigoplus_{i=1}^{k} M_i$ be a direct sum of R-modules. Then*

$$\operatorname{Hom}_R(M, M) \cong \big(\operatorname{Hom}_R(M_j, M_i)\big)_{1 \leq i,j \leq k}$$

as rings, where the right-hand side is a ring T of $k \times k$ matrices under the usual matrix addition and multiplication, with (i, j) entry in $\operatorname{Hom}_R(M_j, M_i)$.

Proof. Let $\lambda_j : M_j \to M$ be the natural inclusion and $\pi_i : M \to M_i$ the natural projection. Two key identities hold:

$$\sum_{\ell=1}^{k} \lambda_\ell \pi_\ell = \mathrm{id}_M, \qquad \pi_p \lambda_q = \delta_{pq} \, \mathrm{id}_{M_p}.$$

The first says every $x \in M$ equals the sum of its components; the second says projecting the q-th inclusion gives zero unless $p = q$.

Define $\sigma : \operatorname{Hom}_R(M, M) \to T$ by $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$.

Ring homomorphism. Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left(\sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi \, \mathrm{id}_M \, \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

Injective. Suppose $\sigma(\phi) = 0$, so $\pi_i \phi \lambda_j = 0$ for all i, j. Applying λ_i on the left and summing over i using the first identity:

$$\phi \lambda_j = \mathrm{id}_M \cdot \phi \lambda_j = \Big(\sum_i \lambda_i \pi_i \Big) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying π_j on the right and summing over j:

$$\phi = \phi \cdot \mathrm{id}_M = \phi \Big(\sum_j \lambda_j \pi_j \Big) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

Surjective. Given $f = (f_{ij}) \in T$, set $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$. Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i \, f_{ij} \, \pi_j \lambda_t = \sum_{i,j} \delta_{si} \, f_{ij} \, \delta_{jt} = f_{st}.$$

Hence $\sigma(\phi) = f$, so σ is surjective. □

Definition 9 (Quotient module)**.** *Let N be an R-submodule of an R-module M. The set of cosets $M/N = \{a + N \mid a \in M\}$ becomes an R-module under*

$$(a + N) + (b + N) = (a + b) + N, \qquad r(a + N) = ra + N.$$

It is called the quotient module *of M by N.*

Theorem 22 (4.31 — Correspondence Theorem for Modules)**.** *Let N be an R-submodule of an R-module M. Every R-submodule of M/N is of the form U/N, where U is an R-submodule of M containing N.*

Proof. The canonical projection $f : M \to M/N$, $f(x) = x + N$, is an R-homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \qquad f(rx) = rx+N = r(x+N) = rf(x).$$

Let X be an R-submodule of M/N. The structure of the proof is: define U as the preimage of X under f, show U is a submodule of M containing N, and conclude $X = U/N$.

Set $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$.

$N \subseteq U$. For any $x \in N$, $f(x) = x+N = 0+N = 0_{M/N} \in X$ (since X is a submodule, it contains 0). Hence $x \in U$.

U is an R-submodule. Since $0 \in N \subseteq U$, U is non-empty. For $x, y \in U$ and $r \in R$, since f is an R-homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \qquad f(rx) = rf(x) \in X,$$

so $x - y \in U$ and $rx \in U$.

$X = U/N$. By definition of U, $f(U) \subseteq X$. For the reverse inclusion, take any $x \in X$; since f is surjective there exists $y \in M$ with $f(y) = x$, so $y \in U$ and $x \in f(U)$. Hence $X = f(U) = U/N$. □

Theorem 23 (4.32 — First Isomorphism Theorem for Modules)**.** *Let $f : M \to N$ be an R-homomorphism. Then $M/\ker f \cong f(M)$.*

Proof. Define $g : M/\ker f \to f(M)$ by $g(x + \ker f) = f(x)$. We know from group theory that g is an isomorphism of abelian groups (well-defined and injective because $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$; surjective onto $f(M)$ by definition). It remains to check that g respects the R-action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f).$$ □

Theorem 24 (Second Isomorphism Theorem for Modules)**.** *Let U and V be R-submodules of an R-module M. Then $(U + V)/V \cong U/(U \cap V)$.*

Proof. Define $f : U \to (U + V)/V$ by $f(u) = u + V$.

f is an R-homomorphism. For $u_1, u_2 \in U$ and $r \in R$:

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

f is surjective. Any element of $(U + V)/V$ has the form $(u + v) + V = u + V = f(u)$ for some $u \in U$, $v \in V$ (since $v \in V$ means $v + V = V$).

$\ker f = U \cap V$. We have $f(u) = 0$ iff $u + V = V$ iff $u \in V$, so $\ker f = \{u \in U \mid u \in V\} = U \cap V$.

By the First Isomorphism Theorem, $U/(U \cap V) \cong (U + V)/V$. □

Theorem 25 (Third Isomorphism Theorem for Modules)**.** *Let $U \subseteq V$ be R-submodules of an R-module M. Then $(M/U)/(V/U) \cong M/V$.*

Proof. Define $f : M/U \to M/V$ by $f(x+U) = x+V$.

f is well-defined. If $x+U = y+U$ then $x-y \in U \subseteq V$, so $x+V = y+V$.

f is an R-homomorphism. For $x+U, y+U \in M/U$ and $r \in R$:

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

f is surjective. For any $x+V \in M/V$, $f(x+U) = x+V$.

$\ker f = V/U$. We have $f(x+U) = 0$ iff $x+V = V$ iff $x \in V$, so $\ker f = \{x+U \mid x \in V\} = V/U$.

By the First Isomorphism Theorem, $(M/U)/(V/U) \cong M/V$. □

Theorem 26 (4.33)**.** *Let $M_1, \ldots, M_n$ be R-modules and $A_i \leq M_i$ an R-submodule for each i. Then*

$$\frac{M_1 \times \cdots \times M_n}{A_1 \times \cdots \times A_n} \cong \frac{M_1}{A_1} \times \cdots \times \frac{M_n}{A_n}.$$

Since the product is finite, $\times$ and $\oplus$ agree here, so equivalently $(M_1 \oplus \cdots \oplus M_n)/(A_1 \oplus \cdots \oplus A_n) \cong M_1/A_1 \oplus \cdots \oplus M_n/A_n$.

Proof. Define $f : M_1 \times \cdots \times M_n \to M_1/A_1 \times \cdots \times M_n/A_n$ by

$$f(m_1, \ldots, m_n) = (m_1 + A_1, \ldots, m_n + A_n).$$

f is an R-homomorphism. Each coordinate is the quotient map $\pi_i : M_i \to M_i/A_i$, and operations in a product are coordinatewise, so f respects addition and scalar multiplication because each π_i does.

f is surjective. Any element of the target is $(m_1+A_1, \ldots, m_n+A_n)$ for some $m_i \in M_i$, and this equals $f(m_1, \ldots, m_n)$.

$\ker f = A_1 \times \cdots \times A_n$. We have $f(m_1, \ldots, m_n) = 0$ iff $m_i + A_i = A_i$ for every i, iff $m_i \in A_i$ for every i.

By the First Isomorphism Theorem, $(M_1 \times \cdots \times M_n)/(A_1 \times \cdots \times A_n) \cong M_1/A_1 \times \cdots \times M_n/A_n$. □

Lecture 6 — Simple, Completely Reducible, and Free Modules

Definition 10 (5.1 — Simple module)**.** *Let R be a ring with unity. An R-module M is called* simple *(or* irreducible*) if $M \neq 0$ and the only R-submodules of M are 0 and M.*

Theorem 27 (5.5 — Characterisation of Simple Modules)**.** *Let R be a ring with unity and M an R-module. The following are equivalent:*

(i) M is simple;

(ii) $M \neq 0$, and M is generated by any $0 \neq x \in M$;

(iii) $M \cong R/I$, where I is a maximal left ideal of R.

Proof. (i) $\Rightarrow$ (ii): Let $0 \neq x \in M$. Then Rx is a non-zero submodule of M, so $Rx = M$ by simplicity.

(ii) $\Rightarrow$ (iii): Take $0 \neq x \in M$ and define $f : R \to M$ by $f(r) = rx$. This is a surjective R-homomorphism, so letting $I = \ker f$, the First Isomorphism Theorem gives $R/I \cong M$. Note that (ii) implies M is simple: if $0 \neq N \leq M$ and $0 \neq x \in N$, then $M = Rx \subseteq N$. Hence $R/I \cong M$ is simple. By the Correspondence Theorem (4.31), any left ideal J with $I \subseteq J \subseteq R$ corresponds to a submodule J/I of R/I. Simplicity forces $J/I = 0$ or $J/I = R/I$, i.e. $J = I$ or $J = R$, so I is maximal.

(iii) $\Rightarrow$ (i): Let $0 \neq N \leq M \cong R/I$. By the Correspondence Theorem (4.31), $N \cong J/I$ for some left ideal J with $I \subsetneq J \subseteq R$. Since I is maximal, $J = R$, so $N = M$. □

Theorem 28 (5.6 — Schur's Lemma)**.** *Let M be a simple R-module. Then* $\mathrm{Hom}_R(M, M)$ *is a division ring.*

Proof. Let $0 \neq f \in \mathrm{Hom}_R(M, M)$. We show f is invertible and $f^{-1} \in \mathrm{Hom}_R(M, M)$.

f is bijective. By Lemma 3, $\ker f$ and $\operatorname{im} f$ are submodules of M, so simplicity forces each to be 0 or M. As $f \neq 0$, we have $\ker f \neq M$ and $\operatorname{im} f \neq 0$, so $\ker f = 0$ (injective) and $\operatorname{im} f = M$ (surjective).

f^{-1} is an R-homomorphism. We cannot push f^{-1} through operations directly, but we can push f through them. The strategy is: to prove $f^{-1}(u) = v$, it suffices to show $f(u) = f(v)$, since f is injective. For $x, y \in M$ and $r \in R$:

$$f\big(f^{-1}(x+y)\big) = x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f\big(f^{-1}(x) + f^{-1}(y)\big),$$
$$f\big(f^{-1}(rx)\big) = rx = rf(f^{-1}(x)) = f\big(rf^{-1}(x)\big),$$

where we used that f is itself an R-homomorphism in the last step of each line. Injectivity of f gives $f^{-1}(x+y) = f^{-1}(x) + f^{-1}(y)$ and $f^{-1}(rx) = rf^{-1}(x)$. □

Corollary 5 (Non-isomorphic simple modules)**.** *Let M and N be simple R-modules. If $M \ncong N$, then*

$$\mathrm{Hom}_R(M, N) = 0.$$

Proof. Let $f : M \to N$ be a nonzero R-homomorphism. By Lemma 3, $\ker f \leq M$ and $\operatorname{im} f \leq N$. Since $\ker f$ is a submodule of the simple module M and $f \neq 0$, we have $\ker f = 0$. Since $\operatorname{im} f$ is a nonzero submodule of the simple module N, we have $\operatorname{im} f = N$. Thus f is an isomorphism, contradicting $M \ncong N$. Therefore every homomorphism $M \to N$ is zero, so $\mathrm{Hom}_R(M, N) = 0$. □

Definition 11 (5.7 — Completely reducible module)**.** *Let R be a ring with unity. An R-module M is called* completely reducible *(or* semisimple*) if $M = \sum_{\alpha \in \Lambda} M_\alpha$, where each M_α is a simple R-submodule.*

Theorem 29 (5.9)**.** *Let $M = \sum_{\alpha \in \Lambda} M_\alpha$ be a sum of simple R-submodules, and let K be a submodule of M. Then there exists $\Lambda' \subseteq \Lambda$ such that $M = K \oplus \big(\bigoplus_{\alpha \in \Lambda'} M_\alpha\big)$.*

Proof. Let S be the family of subsets $A \subseteq \Lambda$ satisfying two conditions: $\bigoplus_{\alpha \in A} M_\alpha$ is a direct sum, and $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$. Together these mean $K \oplus \bigoplus_{\alpha \in A} M_\alpha$ is a genuine direct sum. Since $\emptyset$ satisfies both conditions trivially, $S \neq \emptyset$. Partially order S by inclusion; every chain (A_i) in S has upper bound $\bigcup A_i$ in S. By Zorn's Lemma, S has a maximal element A.

Set $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$. We claim $N = M$. Fix any $\beta \in \Lambda$. Since M_β is simple, $M_\beta \cap N$ is either 0 or M_β. If $M_\beta \cap N = 0$, then M_β intersects neither K nor $\bigoplus_{\alpha \in A} M_\alpha$, so $A \cup \{\beta\}$ would satisfy both conditions in S, contradicting maximality of A. Hence $M_\beta \cap N = M_\beta$, i.e. $M_\beta \subseteq N$, for every $\beta \in \Lambda$. Since $M = \sum_\beta M_\beta$, we conclude $N = M$. □

Corollary 6 (5.10)**.** *If $M = \sum_{\alpha \in \Lambda} M_\alpha$ is a sum of simple R-submodules, then there exists $\Lambda' \subseteq \Lambda$ such that $M = \bigoplus_{\alpha \in \Lambda'} M_\alpha$.*

Proof. Apply Theorem 5.9 with $K = \{0\}$. □

Definition 12 (5.11 — Linear independence and basis)**.** *A sequence $x_1, \ldots, x_n$ in an R-module M is* linearly independent *if $\sum_{i=1}^n a_i x_i = 0$ implies $a_1 = \cdots = a_n = 0$ for any $a_i \in R$. A subset $S \subseteq M$ is linearly independent if every finite sequence of distinct elements of S is linearly independent.*

A subset $B \subseteq M$ is a basis *of M if B generates M and B is linearly independent. An R-module that has a basis is called a* free module*.*

Lecture 7 — Free Modules and Chain Conditions

Theorem 30 (5.19)**.** *Let M be a free R-module with basis $\{f_1, \ldots, f_n\}$. Then $M \cong R^n$ as R-modules.*

Proof. Define $\varphi : M \to R^n$ by $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \ldots, r_n)$, i.e. read off the coordinates in the basis. Since every element of M has a *unique* representation as $\sum r_i f_i$ (by linear independence of the f_i), φ is well-defined. It is an R-homomorphism: $\varphi(r \cdot \sum s_i f_i) = \varphi(\sum (r s_i) f_i) = (r s_1, \ldots, r s_n) = r \cdot (s_1, \ldots, s_n) = r \cdot \varphi(\sum s_i f_i)$, where the last scalar multiplication is coordinate-wise in R^n. It is injective (if $(r_1, \ldots, r_n) = 0$ then each $r_i = 0$, so $m = 0$), and surjective (every $(r_1, \ldots, r_n) \in R^n$ is the image of $\sum r_i f_i$). □

Theorem 31 (5.20)**.** *Let M be a finitely generated free R-module. Then every basis of M is finite.*

Proof. Let $\{e_i\}_{i \in \Lambda}$ be a basis and $\{x_1, \ldots, x_n\}$ a finite generating set. Since elements of a module are finite sums by definition, each x_j involves only finitely many nonzero coefficients, so is supported on a finite subset $S_j \subseteq \Lambda$. Every basis element must appear in the expression of some generator (otherwise it could not be reached), so $\Lambda \subseteq S_1 \cup \cdots \cup S_n$, a finite union of finite sets. Hence Λ is finite. □

Theorem 32 (5.21)**.** *Let M be a finitely generated free module over a commutative ring R. Then all bases of M have the same number of elements.*

Proof. Suppose M has two bases of sizes m and n respectively. By Theorem 5.19, $M \cong R^m$ and $M \cong R^n$, so $R^m \cong R^n$. It suffices to show $m = n$.

Assume for contradiction $m < n$. Let $\phi : R^m \to R^n$ be an R-isomorphism with inverse $\psi = \phi^{-1}$. Let $(e_1, \ldots, e_m)$ and $(f_1, \ldots, f_n)$ be the standard bases of R^m and R^n. Since ϕ is an R-homomorphism, it is determined by where it sends each e_i, and each image lands in R^n, so we can write

$$\phi(e_i) = \sum_{j=1}^{n} a_{ji} f_j, \qquad \psi(f_j) = \sum_{k=1}^{m} b_{kj} e_k.$$

Let $A = (a_{ji})$ be the resulting $n \times m$ matrix and $B = (b_{kj})$ the $m \times n$ matrix. Since $\psi\phi = \mathrm{id}_{R^m}$ and $\phi\psi = \mathrm{id}_{R^n}$, substituting the expressions above and using linear independence of the bases gives $BA = I_m$ and $AB = I_n$.

Pad these to $n \times n$ matrices by setting $A' = \begin{pmatrix} A & 0 \end{pmatrix}$ and $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$. Then $A'B' = I_n$, so $\det(A'B') = 1$. But $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$, so $\det(B'A') = 0$. Since R is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A')\det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry $n < m$ also fails, so $m = n$. □

Definition 13 (5.22 — Rank)**.** *The number of elements in any basis of a finitely generated free module M over a commutative ring R is called the* rank *of M, written* $\mathrm{rk}(M)$*. If R is a field, the rank is called the* dimension*.*

Lemma 4 (5.24)**.** *Every finitely generated R-module is a homomorphic image of a finitely generated free module.*

Proof. Let M be generated by $x_1, \ldots, x_n$. Define $\phi : R^n \to M$ by $\phi(r_1, \ldots, r_n) = \sum_{i=1}^n r_i x_i$. This is a surjective R-homomorphism (surjective since the x_i generate M). The generators x_i may satisfy relations, so ϕ need not be injective — but by the First Isomorphism Theorem $M \cong R^n / \ker\phi$, expressing M as a quotient of the free module R^n. □

Lemma 5 (5.25)**.** *Let R be a PID and F a free R-module of rank n. Then every submodule $K \leq F$ is free of rank $m \leq n$.*

Proof. By Theorem 5.19, $F \cong R^n$, so we work with $K \leq R^n$ and induct on n. The base case $n = 0$ is trivial since $K = 0$ is free on the empty set.

Define $\pi : R^n \to R$ by $\pi(x_1, \ldots, x_n) = x_1$, and restrict it to K.

Case 1: $\pi|_K = 0$. Every element of K has first coordinate zero, so $K \leq \{0\} \times R^{n-1} \cong R^{n-1}$. By induction K is free of rank $\leq n - 1 \leq n$.

Case 2: $\pi|_K \neq 0$. The image $\pi(K)$ is a nonzero ideal of R. Since R is a PID, $\pi(K) = Ra$ for some $a \neq 0$. Pick $k \in K$ with $\pi(k) = a$.

We claim $K = Rk \oplus \ker(\pi|_K)$, where $\ker(\pi|_K) = \{x \in K \mid x_1 = 0\}$ is the set of elements of K with first coordinate zero, so $\ker(\pi|_K) \leq \{0\} \times R^{n-1} \cong R^{n-1}$.

$K = Rk + \ker(\pi|_K)$: For any $x \in K$, write $\pi(x) = ba$ for some $b \in R$. Then $\pi(x - bk) = ba - ba = 0$, so $x - bk \in \ker(\pi|_K)$, and $x = bk + (x - bk)$.

The sum is direct: If $ck \in \ker(\pi|_K)$, then $0 = \pi(ck) = ca$. Since R is a PID (hence an integral domain) and $a \neq 0$, we get $c = 0$.

By induction $\ker(\pi|_K)$ is free of rank $s \leq n - 1$, and $Rk \cong R$ is free of rank 1. Hence $K \cong R^s \oplus R = R^{s+1}$, which has rank $m = s + 1 \leq n$. □

Lecture 8 — Noetherian and Artinian Modules

Definition 14 (6.1 — Noetherian and Artinian modules)**.** *An R-module M is called* Noetherian *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilises, i.e. there exists k such that $M_k = M_{k+1} = \cdots$. This is called the ascending chain condition *(ACC). An R-module M is called* Artinian *if every descending chain of submodules stabilises (*descending chain condition*, DCC).*

Theorem 33 (6.5)**.** *Let M be an R-module. The following are equivalent:*

(i) M is Noetherian;

(ii) every submodule of M is finitely generated;

(iii) every non-empty family S of submodules of M has a maximal element.

Proof. (i) $\Rightarrow$ (ii): Let $N \leq M$ and suppose N is not finitely generated. For any finite choice $a_1, \ldots, a_k \in N$ we have $N \neq (a_1, \ldots, a_k)$, so we can always pick $a_{k+1} \in N \setminus (a_1, \ldots, a_k)$. This produces an infinite strictly ascending chain $(a_1) \subsetneq (a_1, a_2) \subsetneq \cdots$, contradicting that M is Noetherian.

(ii) $\Rightarrow$ (iii): Suppose S has no maximal element. Then starting from any $N_0 \in S$ we can always find a strictly larger member, producing an infinite strictly ascending chain $N_0 \subsetneq N_1 \subsetneq N_2 \subsetneq \cdots$. Let $N = \bigcup_i N_i$. For $x, y \in N$ and $r \in R$, both x and y lie in some N_ℓ (take the larger index), so $x - y, rx \in N_\ell \subseteq N$; hence N is a submodule. By (ii), $N = (a_1, \ldots, a_n)$ for some $a_j \in N$. Each a_j lies in some N_{u_j}, so all generators lie in N_μ for $\mu = \max\{u_1, \ldots, u_n\}$. But then $N = (a_1, \ldots, a_n) \subseteq N_\mu \subseteq N$, so $N_\mu = N$, contradicting that the chain is strictly ascending.

(iii) $\Rightarrow$ (i): Given an ascending chain $M_1 \subseteq M_2 \subseteq \cdots$, by (iii) the family $\{M_i\}$ has a maximal element M_k. Then $M_k = M_{k+1} = \cdots$, so the chain stabilises. $\square$

Definition 15 (6.6 — Finitely cogenerated)**.** *An R-module M is* finitely cogenerated *if for every family $(M_\alpha)_{\alpha \in \Lambda}$ of submodules of M,*

$$\bigcap_{\alpha \in \Lambda} M_\alpha = 0 \implies \bigcap_{\alpha \in \Lambda'} M_\alpha = 0$$

for some finite subset $\Lambda' \subseteq \Lambda$.

Theorem 34 (6.8)**.** *Let M be an R-module. The following are equivalent:*

(i) M is Artinian;

(ii) every quotient module of M is finitely cogenerated;

(iii) every non-empty family S of submodules of M has a minimal element.

Proof. (i) $\Rightarrow$ (ii): Let $N \leq M$ and take any family $(M_\alpha/N)_{\alpha \in \Lambda}$ of submodules of M/N with $\bigcap_{\alpha \in \Lambda} M_\alpha/N = 0$, i.e. $\bigcap_{\alpha \in \Lambda} M_\alpha = N$. Consider the family of finite intersections $\left\{\bigcap_{\alpha \in \Lambda'} M_\alpha \mid \Lambda' \subseteq \Lambda \text{ finite}\right\}$. As we add more submodules to the intersection it can only get smaller, so this is a descending family of submodules of M. Since M is Artinian every descending chain stabilises, so there exists a finite $\Lambda' \subseteq \Lambda$ such that

$$\bigcap_{\alpha \in \Lambda'} M_\alpha = \bigcap_{\alpha \in \Lambda} M_\alpha = N.$$

Hence $\bigcap_{\alpha \in \Lambda'} M_\alpha/N = 0$, so M/N is finitely cogenerated.

(ii) $\Rightarrow$ (iii): Suppose S has no minimal element. Then starting from any $N_0 \in S$ we can always find a strictly smaller member, producing an infinite strictly descending chain $N_0 \supsetneq N_1 \supsetneq N_2 \supsetneq \cdots$. Let $N = \bigcap_i N_i$, which is a submodule of M. By (ii), M/N is finitely cogenerated. The family (N_i/N) of submodules of M/N satisfies $\bigcap_i N_i/N = N/N = 0$, so by finite cogeneration there exists a finite subset, say $\{N_0, \ldots, N_k\}$, with $\bigcap_{i=0}^{k} N_i/N = 0$, i.e. $\bigcap_{i=0}^{k} N_i = N$. Since the chain is descending, $\bigcap_{i=0}^{k} N_i = N_k$, so $N_k = N \subseteq N_{k+1} \subsetneq N_k$, a contradiction.

(iii) $\Rightarrow$ (i): Given a descending chain $M_1 \supseteq M_2 \supseteq \cdots$, by (iii) the family $\{M_i\}$ has a minimal element M_k. Then $M_k = M_{k+1} = \cdots$, so the chain stabilises. □

Theorem 35 (6.9)**.** *Let M be a Noetherian (respectively Artinian) R-module and $K \leq M$ a submodule. Then K and M/K are both Noetherian (respectively Artinian).*

Proof. Both cases at once, using chains. A chain of submodules of K is also a chain in M, so it stabilises; hence K is Noetherian (respectively Artinian). For M/K: by Theorem 4.31 (Correspondence Theorem), a chain $L_1/K, L_2/K, \ldots$ of submodules of M/K (ascending, respectively descending) lifts to a chain $L_1, L_2, \ldots$ in M of the same kind, with $K \subseteq L_i$. This chain stabilises, so the chain in M/K stabilises too. □

Lemma 6 (Key lemma for 6.10)**.** *Let $N, K', K \leq M$ be submodules with $K' \subseteq K$. If*

$$K' \cap N = K \cap N \quad \text{and} \quad K' + N = K + N,$$

then $K' = K$. In words: a submodule is determined by its intersection with N and its image modulo N, among comparable submodules.

Proof. Take $x \in K$. Since $x \in K + N = K' + N$, write $x = y + n$ with $y \in K'$ and $n \in N$. Then $n = x - y \in K$, because $y \in K' \subseteq K$. Hence $n \in K \cap N = K' \cap N \subseteq K'$, so $x = y + n \in K'$. Thus $K \subseteq K'$, and $K' = K$. □

Note: the inclusion $K' \subseteq K$ cannot be dropped. In $M = \mathbb{R}^2$ with N the x-axis, two distinct lines K, K' through the origin (both $\neq N$) have $K \cap N = K' \cap N = 0$ and $K + N = K' + N = \mathbb{R}^2$. In a chain, however, any two members are comparable.

Theorem 36 (6.10)**.** *Let M be an R-module and $N \leq M$ a submodule. Then M is Noetherian (respectively Artinian) if and only if both N and M/N are Noetherian (respectively Artinian).*

Proof. ($\Rightarrow$) This is Theorem 6.9.

($\Leftarrow$) Both cases at once. Assume N and M/N are Noetherian (respectively Artinian), and let

$$K_1 \subseteq K_2 \subseteq \cdots \qquad (\text{respectively } K_1 \supseteq K_2 \supseteq \cdots)$$

be a chain of submodules of M. Consider the two induced chains, of the same kind:

$$K_1 \cap N,\ K_2 \cap N,\ \ldots \quad \text{in } N, \qquad \frac{K_1 + N}{N},\ \frac{K_2 + N}{N},\ \ldots \quad \text{in } M/N.$$

Both stabilise by assumption, so there is q such that for all $i \geq q$,

$$K_i \cap N = K_q \cap N \quad \text{and} \quad \frac{K_i + N}{N} = \frac{K_q + N}{N}.$$

Since $K_i + N$ and $K_q + N$ both contain N, the Correspondence Theorem (4.31) turns the second equality into $K_i + N = K_q + N$.

For $i \geq q$, the modules K_i and K_q are comparable: $K_q \subseteq K_i$ in the ascending case, and $K_i \subseteq K_q$ in the descending case. The two hypotheses of the Key Lemma are symmetric in K_i and K_q, so applying it with the smaller of the two as K' gives $K_i = K_q$. Hence the chain stabilises, and M is Noetherian (respectively Artinian). □

Corollary 7 (6.11)**.** *If $M_1, \ldots, M_n$ are Noetherian (respectively Artinian) R-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (respectively Artinian).*

Proof. By induction it suffices to treat the case of two modules M and N. By Theorem 6.10, to show $M \oplus N$ is Noetherian (respectively Artinian) it suffices to find a single submodule of $M \oplus N$ such that both the submodule and its quotient module are Noetherian (respectively Artinian). Take $M \oplus 0 \leq M \oplus N$. Then:

- the submodule $M \oplus 0 \cong M$ is Noetherian (respectively Artinian) by assumption;
- the quotient $(M \oplus N)/(M \oplus 0) \cong N$ is Noetherian (respectively Artinian) by assumption.

Hence by Theorem 6.10, $M \oplus N$ is Noetherian (respectively Artinian). □

Definition 16 (6.12 — Noetherian and Artinian rings)**.** *A ring R is called* left Noetherian *(respectively* left Artinian*) if R, regarded as a left R-module, is Noetherian (respectively Artinian). By default, Noetherian (respectively Artinian) ring means left Noetherian (respectively left Artinian).*

Theorem 37 (6.13)**.** *Let R be a ring. The following are equivalent:*

(i) R is Noetherian (respectively Artinian);

(ii) every left ideal A of R is finitely generated (respectively every R/A is finitely cogenerated);

(iii) every non-empty set of left ideals of R has a maximal (respectively minimal) element.

Proof. This is Theorem 6.5 (respectively 6.8) applied to R regarded as a left R-module, where submodules of R are precisely the left ideals of R. □

Theorem 38 (6.19)**.** *Let $R_1, \ldots, R_n$ be Noetherian (respectively Artinian) rings. Then $R_1 \oplus \cdots \oplus R_n$ is Noetherian (respectively Artinian).*

Proof. This follows immediately from Corollary 6.11, viewing each R_i as an R-module. □

Product of ideals. For ideals A and B in a ring R, define

$$AB = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; n \in \mathbb{N},\, a_i \in A,\, b_i \in B \right\}.$$

Then AB is again an ideal of R.

Definition 17 (6.20 — Prime ideal)**.** *Let R be a ring. An ideal $P \neq R$ is* prime *if whenever A and B are ideals in R with $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$.*

(For commutative R this recovers the classical definition: $ab \in P \Rightarrow a \in P$ or $b \in P$.)

Theorem 39 (6.21)**.** *Let R be a ring with unity. Then every maximal ideal is prime.*

Proof. Let M be a maximal ideal and let A, B be ideals with $AB \subseteq M$. Suppose $A \not\subseteq M$. Since M is maximal, the only ideal strictly containing M is R itself, so $A + M = R$. Write $1 = a + m$ for some $a \in A$ and $m \in M$. For any $b \in B$,

$$b = 1 \cdot b = (a + m)b = ab + mb \in AB + MB \subseteq M + M = M.$$

Hence $B \subseteq M$, so M is prime. □

Note: The converse is false: (0) is a prime ideal in $\mathbb{Z}$ but not a maximal ideal.

Lecture 9

Lemma 7 (6.22)**.** *If R is a Noetherian ring, then each ideal contains a finite product of prime ideals.*

Proof. Let $\mathcal{F}$ be the family of ideals of R that do not contain any finite product of prime ideals. We want to show $\mathcal{F} = \emptyset$.

Suppose for a contradiction that $\mathcal{F} \neq \emptyset$. Since R is Noetherian, by Theorem 6.13(iii) the family $\mathcal{F}$ has a maximal element, say A.

We claim A is not a prime ideal. Indeed, any prime ideal P contains the one-factor product P itself, so no prime ideal can belong to $\mathcal{F}$.

Since A is not prime, there exist ideals B and C with $BC \subseteq A$, but $B \not\subseteq A$ and $C \not\subseteq A$. Then $B + A \supsetneq A$ and $C + A \supsetneq A$, so by maximality of A in $\mathcal{F}$, neither $B + A$ nor $C + A$ belongs to $\mathcal{F}$. Hence both contain finite products of prime ideals. But then $(B + A)(C + A)$ also contains a finite product of prime ideals, and

$$(B + A)(C + A) \subseteq BC + BA + AC + A^2 \subseteq A.$$

So A contains a finite product of prime ideals, contradicting $A \in \mathcal{F}$. Hence $\mathcal{F} = \emptyset$. □

Theorem 40 (6.23 — Hilbert's Basis Theorem)**.** *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

Proof. Suppose for a contradiction that $I \subseteq R[x]$ is a left ideal that is not finitely generated. We recursively construct a sequence of polynomials as follows. Set

$$J_0 = (0) \subseteq R[x].$$

Since I is not finitely generated, $I \setminus J_0 \neq \emptyset$. Pick $f_0 \in I \setminus J_0$ of minimal degree (exists since degrees are non-negative integers). For $n \geq 1$, given $f_0, \ldots, f_{n-1}$, let

$$J_n = (f_0, \ldots, f_{n-1}) \subseteq R[x]$$

be the left ideal in $R[x]$ generated by $f_0, \ldots, f_{n-1}$. Since I is not finitely generated, $I \setminus J_n \neq \emptyset$, so we may pick $f_n \in I \setminus J_n$ of minimal degree.

Since $J_n \subsetneq J_{n+1}$, the set $I \setminus J_{n+1} \subsetneq I \setminus J_n$, so $\deg(f_n)$ is non-decreasing in n.

For each n, let $a_n \in R$ denote the leading coefficient of f_n. Consider the ascending chain of left ideals in R:

$$L_1 \subseteq L_2 \subseteq \cdots, \qquad \text{where } L_n = (a_0, \ldots, a_{n-1}) \subseteq R.$$

Since R is Noetherian, this chain stabilises: there exists $N \in \mathbb{N}$ such that $L_N = L_{N+1}$, that is,

$$a_N \in L_N = (a_0, \ldots, a_{N-1}) \subseteq R.$$

So we may write $a_N = \sum_{i<N} u_i a_i$ for some $u_i \in R$. Now define

$$g = \sum_{i<N} u_i x^{\deg(f_N)-\deg(f_i)} f_i \in J_N.$$

The exponents $\deg(f_N) - \deg(f_i) \geq 0$ since degrees are non-decreasing. The leading term of $u_i x^{\deg(f_N)-\deg(f_i)} f_i$ is $u_i a_i x^{\deg(f_N)}$, so the leading term of g is

$$\left(\sum_{i<N} u_i a_i\right) x^{\deg(f_N)} = a_N x^{\deg(f_N)},$$

which equals the leading term of f_N. Hence $f_N - g$ has degree strictly less than $\deg(f_N)$.

Since $f_N \notin J_N$ and $g \in J_N$, we have $f_N - g \notin J_N$. Since $f_N, g \in I$, we have $f_N - g \in I$. Thus $f_N - g \in I \setminus J_N$.

- If $\deg(f_N) > 0$: then $f_N - g \in I \setminus J_N$ has degree strictly less than $\deg(f_N)$, contradicting the minimality of $\deg(f_N)$.
- If $\deg(f_N) = 0$: then all degrees are 0, so $f_N - g = 0$, meaning $f_N = g \in J_N$, contradicting $f_N \in I \setminus J_N$.

In both cases we reach a contradiction, so I must be finitely generated and $R[x]$ is Noetherian. □

Definition 18 (7.1 — Nilpotent elements and ideals)**.** *Let R be a ring.*

- *An element $a \in R$ is* nilpotent *if $a^n = 0$ for some $n \in \mathbb{N}$.*
- *A left ideal I of R is a* nil ideal *if every element of I is nilpotent.*
- *A left ideal I of R is* nilpotent *if $I^n = 0$ for some $n \in \mathbb{N}$, where I^n denotes the set of all finite sums of products $i_1 i_2 \cdots i_n$ with each $i_k \in I$.*

Every nilpotent ideal is nil (if $I^n = 0$ then any $a \in I$ satisfies $a^n \in I^n = 0$), but the converse fails in general.

Theorem 41 (7.5)**.** *If J is a nil left ideal in an Artinian ring R, then J is nilpotent.*

Proof. Suppose for a contradiction that $J^k \neq 0$ for all $k \in \mathbb{N}$. The family $(J, J^2, J^3, \ldots)$ is a descending chain of left ideals of R. Since R is Artinian, this family has a minimal element, say $B = J^m$ for some m. Then

$$B^2 = J^{2m} \subseteq J^m = B,$$

and by minimality of B we have $B^2 = B$.

Now consider the family

$$\mathcal{F} = \{A \mid A \text{ is a left ideal of } R,\ A \subseteq B,\ BA \neq 0\}.$$

Since $B^2 = B \neq 0$, we have $B \in \mathcal{F}$, so $\mathcal{F} \neq \emptyset$. Since R is Artinian, $\mathcal{F}$ has a minimal element A.

Since $BA \neq 0$, there exists $a \in A$ with $Ba \neq 0$. Note that $Ba = \{ba \mid b \in B\}$ is a left ideal contained in A, and

$$B(Ba) = B^2a = Ba \neq 0,$$

so $Ba \in \mathcal{F}$. By minimality of A we conclude $Ba = A$.

Hence there exists $b \in B$ with $ba = a$. Applying b repeatedly gives $b^i a = a$ for all $i \in \mathbb{N}$. But $b \in B \subseteq J$ and J is nil, so $b^n = 0$ for some $n \in \mathbb{N}$. Then $a = b^n a = 0$, contradicting $Ba \neq 0$.

Therefore $J^k = 0$ for some $k \in \mathbb{N}$, i.e. J is nilpotent. □

Lemma 8 (7.6)**.** *Let R be a Noetherian ring. Then the sum of nilpotent left ideals in R is a nilpotent left ideal.*

Proof. Let $B = \sum_{i \in \Lambda} A_i$ be a sum of nilpotent left ideals. Since R is Noetherian, B is finitely generated as a left ideal, say $B = (x_1, \ldots, x_m)$. Each x_j lies in a finite sum of the A_i's, so B is contained in a finite sum $A_1 + \cdots + A_n$ (reindexing if necessary), and hence $B = A_1 + \cdots + A_n$.

It remains to show B is nilpotent. For each i, let $n_i \in \mathbb{N}$ be such that $A_i^{n_i} = 0$. Set $N = n_1 + \cdots + n_n$. Any product of N elements from $B = A_1 + \cdots + A_n$, when expanded, is a sum of products of the form $a_{i_1} a_{i_2} \cdots a_{i_N}$ with each $a_{i_k} \in A_{i_k}$. By pigeonhole, among the N indices $i_1, \ldots, i_N \in \{1, \ldots, n\}$, at least one value j appears at least n_j times. The consecutive factors from A_j multiply to give an element of $A_j^{n_j} = 0$. Hence every such product is 0, so $B^N = 0$. □

Lecture 10

Theorem 42 (7.7)**.** *Let R be a Noetherian ring having no non-zero nilpotent ideals. Then R has no non-zero nil ideals.*

(Proof non-examinable.)

Theorem 43 (7.8)**.** *Let N be a nil ideal in a Noetherian ring R. Then N is nilpotent.*

Proof. Let T be the sum of all nilpotent ideals in R. By Lemma 7.6, T is itself nilpotent, say $T^k = 0$.

We claim R/T has no non-zero nilpotent ideals. Suppose A/T is a nilpotent ideal in R/T, say $(A/T)^m = 0$. Then $(A^m + T)/T = 0$, so $A^m \subseteq T$. Since $T^k = 0$, we get $A^{mk} \subseteq T^k = 0$, so A is nilpotent. Hence $A \subseteq T$, giving $A/T = 0$.

Since R/T is Noetherian (a quotient of a Noetherian ring) and has no non-zero nilpotent ideals, Theorem 7.7 gives that R/T has no non-zero nil ideals. Consider $(N+T)/T$: every element has the form $n + T$ for some $n \in N$ (since $t + T = T$ for $t \in T$), and since N is nil there exists k with $n^k = 0$, giving $(n + T)^k = n^k + T = T$. Hence $(N + T)/T$ is a nil ideal in R/T, so $(N + T)/T = 0$, which means $N \subseteq T$. Since T is nilpotent, N is nilpotent. □

Lemma 9 (7.9)**.** *Let A be a minimal left ideal in a ring R. Then either $A^2 = 0$ or $A = Re$ for some idempotent $e \in R$.*

Proof. Assume $A^2 \neq 0$.

Step 1: Find $e \in A$ with $ea = a$. Since $A^2 \neq 0$, there exists $a \in A$ with $Aa \neq 0$. The set $Aa = \{ca \mid c \in A\}$ is a left ideal contained in A (since A is a left ideal and $a \in A$). By minimality of A, we have $Aa = A$. In particular, there exists $e \in A$ with $ea = a$.

Step 2: Show $e^2 = e$. We compute $(e^2 - e)a = e(ea) - ea = ea - ea = 0$. To conclude $e^2 = e$, define the left annihilator of a in A:

$$\mathrm{Ann}_A(a) := \{c \in A \mid ca = 0\}.$$

This is a left ideal of R contained in A. Since $ea = a \neq 0$, we have $e \notin \mathrm{Ann}_A(a)$, so $\mathrm{Ann}_A(a) \neq A$. By minimality of A, we conclude $\mathrm{Ann}_A(a) = 0$. Since $e^2 - e \in A$ and $(e^2 - e)a = 0$, we get $e^2 - e \in \mathrm{Ann}_A(a) = 0$, hence $e^2 = e$.

Step 3: Show $A = Re$. Since $e \in A$ and A is a left ideal, $Re \subseteq A$. Since $e \neq 0$ (as $ea = a \neq 0$), we have $Re \neq 0$. By minimality of A, $Re = A$. □

Theorem 44 (7.10 — Wedderburn–Artin)**.** *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then*

$$R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$$

for some $n_1, \ldots, n_k \in \mathbb{N}$ and division rings $D_1, \ldots, D_k$.

Proof. We build the result from two claims, then combine them.

Claim 1. *Each non-zero left ideal A of R is of the form Re for some idempotent $e \in R$.*

By Lemma 7.9, every minimal left ideal is either nilpotent or of the form Re for some idempotent e. Since R has no non-zero nilpotent ideals, every minimal left ideal is of the form Re.

Now let A be any non-zero left ideal. Since R is Artinian, A contains a minimal left ideal, hence a non-zero idempotent. Thus the family

$$\mathcal{F} = \{R(1 - f) \cap A \mid f \text{ a non-zero idempotent in } A\}$$

is non-empty, and by the Artinian condition has a minimal member $R(1 - e) \cap A$.

Subclaim *(non-examinable).* $R(1 - e) \cap A = 0$.

Invoking the subclaim: for any $a \in A$, $a(1 - e) \in R(1 - e) \cap A = 0$, so $a = ae \in Re$. Hence $A \subseteq Re \subseteq A$, giving $A = Re$.

Claim 2. *$R = Re_1 \oplus \cdots \oplus Re_n$ is a finite direct sum of minimal left ideals.*

Let S be the sum of all minimal left ideals in R; since R is Artinian it has at least one minimal left ideal, so $S \neq 0$. By Claim 1, $S = Re$ for some idempotent e.

Suppose $R(1 - e) \neq 0$. Then $R(1 - e)$ is a non-zero left ideal, so by the Artinian condition it contains a minimal left ideal A. But $A \subseteq S = Re$, giving $A \subseteq Re \cap R(1 - e)$. For any $x \in Re \cap R(1 - e)$, write $x = ae = b(1 - e)$ for some $a, b \in R$. Then $xe = ae^2 = ae = x$, and also $xe = b(1 - e)e = b(e - e^2) = 0$, so $x = 0$. Hence $Re \cap R(1 - e) = 0$, giving $A = 0$, a contradiction.

Therefore $R(1 - e) = 0$. Since R has unity, $1 - e = 0$, i.e. $e = 1$, so $S = R$.

Thus R is the sum of all its minimal left ideals. By Theorem 5.9 there exists a subfamily $(A_i)_{i\in\Lambda'}$ such that $R = \bigoplus_{i\in\Lambda'} A_i$. To see this is finite, write $1 = e_{i_1} + \cdots + e_{i_n}$ as a finite sum with $e_{i_j} \in A_{i_j}$. Then for any $r \in R$, $r = r \cdot 1 = re_{i_1} + \cdots + re_{i_n}$, so $R = Re_{i_1} \oplus \cdots \oplus Re_{i_n}$. After reindexing, $R = Re_1 \oplus \cdots \oplus Re_n$.

Conclusion. The plan: compute $\mathrm{End}_R(R) = \mathrm{Hom}_R(R, R)$ as a matrix ring using Theorem 4.30, use Schur's Lemma to see that it is block diagonal, and finally recover R from $\mathrm{End}_R(R)$.

Step 1: Group isomorphic summands. By Claim 2, $R = Re_1 \oplus \cdots \oplus Re_n$ as left R-modules, where each Re_i is a minimal left ideal, i.e. a simple left R-module. Sort the summands into isomorphism classes; say there are k classes, of sizes $n_1, \ldots, n_k$, and choose a representative N_j of the j-th class. Thus $N_i \ncong N_j$ for $i \neq j$, and every Re_ℓ is isomorphic to exactly one N_j. The order of summands in a direct sum is irrelevant, so we may reorder them so that each class is adjacent. Replacing each summand by its (isomorphic) representative then gives an isomorphism of left R-modules

$$R \;\cong\; M := \underbrace{(N_1 \oplus \cdots \oplus N_1)}_{n_1} \oplus \cdots \oplus \underbrace{(N_k \oplus \cdots \oplus N_k)}_{n_k}.$$

A module isomorphism $\theta : R \to M$ induces a ring isomorphism $\mathrm{End}_R(R) \to \mathrm{End}_R(M)$, $\varphi \mapsto \theta\varphi\theta^{-1}$. Hence $\mathrm{End}_R(R) \cong \mathrm{End}_R(M)$, and it suffices to compute $\mathrm{End}_R(M)$.

Step 2: Apply Schur. Since each N_j is simple, Schur's Lemma 5.6 says every non-zero homomorphism between N_i and N_j is an isomorphism. Therefore

$$\mathrm{Hom}_R(N_i, N_j) = 0 \quad (i \neq j), \qquad D_j := \mathrm{End}_R(N_j) \text{ is a division ring.}$$

Step 3: Compute $\mathrm{End}_R(M)$. View M as a direct sum of $n = n_1 + \cdots + n_k$ simple summands. By Theorem 4.30, $\mathrm{End}_R(M)$ is isomorphic to the ring of $n \times n$ matrices whose (p, q) entry lies in $\mathrm{Hom}_R(q\text{-th summand}, p\text{-th summand})$. By Step 2, this entry is 0 if the two summands are copies of different N_i, N_j, and is an arbitrary element of D_j if both are copies of N_j. Since each class is adjacent, these are exactly the block-diagonal matrices

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_k \end{pmatrix}, \qquad A_j \in M_{n_j}(D_j).$$

Block-diagonal matrices add and multiply block by block, $\mathrm{diag}(A_1, \ldots, A_k)\,\mathrm{diag}(B_1, \ldots, B_k) = \mathrm{diag}(A_1B_1, \ldots, A_kB_k)$, so sending a matrix to its tuple of blocks is a ring isomorphism. Hence

$$\mathrm{End}_R(R) \;\cong\; \mathrm{End}_R(M) \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

Step 4: Identify $\mathrm{End}_R(R)$ *with* R^{op}. Every $\varphi \in \mathrm{End}_R(R)$ is determined by $b := \varphi(1)$, since $\varphi(r) = \varphi(r \cdot 1) = r\varphi(1) = rb$. So every endomorphism is right multiplication $\rho_b : r \mapsto rb$ for a unique $b \in R$, and conversely each ρ_b is an endomorphism. Composition reverses the order of multiplication: $(\rho_b \circ \rho_c)(r) = rcb$, so $\rho_b \circ \rho_c = \rho_{cb} = \rho_{b \cdot_{\mathrm{op}} c}$. Hence $b \mapsto \rho_b$ is a ring isomorphism $R^{\mathrm{op}} \cong \mathrm{End}_R(R)$, and combining with Step 3,

$$R^{\mathrm{op}} \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

Step 5: Pass to opposite rings. Taking opposites of both sides (using $(R^{\mathrm{op}})^{\mathrm{op}} = R$), and noting that the opposite of a direct sum is the direct sum of the opposites,

$$R \;\cong\; M_{n_1}(D_1)^{\mathrm{op}} \oplus \cdots \oplus M_{n_k}(D_k)^{\mathrm{op}}.$$

Transposition gives $M_n(D)^{\mathrm{op}} \cong M_n(D^{\mathrm{op}})$, and D^{op} is again a division ring. Relabelling D_j^{op} as D_j, we conclude

$$R \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k). \qquad \square$$

Theorem 45 (7.11)**.** *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then R is also right and left Artinian, and right and left Noetherian.*

(Non-examinable. Follows from Wedderburn–Artin and the fact that matrix rings over division rings are Noetherian and Artinian on both sides.)

Theorem 46 (7.13 — Maschke)**.** *If G is a finite group, then*

$$\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \oplus \cdots \oplus M_{n_k}(\mathbb{C})$$

for some $n_1, \ldots, n_k \in \mathbb{N}$.

(Non-examinable. Follows from Wedderburn–Artin: one shows $\mathbb{C}[G]$ has no non-zero nilpotent ideals and is Artinian, then uses that the only division ring containing $\mathbb{C}$ in its centre as an algebraically closed field is $\mathbb{C}$ itself.)

Theorem 47 (7.14)**.** *Let R be a ring with unity. The following are equivalent:*

(i) R is left Artinian with no non-zero nilpotent ideals;

(ii) R is left Artinian with no non-zero nil ideals;

(iii) R is a finite direct sum of minimal left ideals;

(iv) every left ideal of R is of the form Re for some idempotent e;

(v) R is a finite direct sum of matrix rings over division rings.

(Non-examinable.)

Definition 19 (7.15 — Semisimple Artinian ring)**.** *A ring satisfying any of the equivalent conditions of Theorem 7.14 is called a* semisimple Artinian ring*.*

Lecture 11 — Smith Normal Form over a PID

Throughout this chapter R is a principal ideal domain with unity.

Definition 20 (8.1 — Row module, column module)**.** *Let $A \in M_{m\times n}(R)$. The submodule of R^n generated by the m rows of A is called the* row module *of A, denoted $\mathcal{R}(A)$. The submodule of R^m generated by the n columns of A is called the* column module *of A, denoted $\mathcal{C}(A)$.*

Since $\mathcal{R}(A)$ and $\mathcal{C}(A)$ are finitely generated submodules of the free modules R^n and R^m respectively, both are free by Lemma 5.25.

Definition 21 (8.2 — Row rank, column rank)**.** *Let $A \in M_{m\times n}(R)$. The rank of $\mathcal{R}(A)$ is called the* row rank *of A, and the rank of $\mathcal{C}(A)$ is called the* column rank *of A.*

Theorem 48 (8.3)**.** *Let $A \in M_{m\times n}(R)$ and let P, Q be invertible $m \times m$ and $n \times n$ matrices over R respectively. Then row rank and column rank of PAQ equal those of A.*

(Proof non-examinable.)

Definition 22 (8.4/8.6 — Elementary operations and matrices)**.** *Let $A \in M_{m\times n}(R)$. There are three* elementary row operations*, each achieved by left-multiplying A by an invertible* elementary matrix *(column operations are achieved by right-multiplying):*

(i) Swap rows i and j: multiply on the left by $S_{ij} = I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$. Here $S_{ij}^{-1} = S_{ij}$.

(ii) Scale row i by invertible $\alpha \in R$: multiply on the left by $M_i(\alpha) = I + (\alpha - 1)E_{ii}$. Here $M_i(\alpha)^{-1} = M_i(\alpha^{-1})$.

(iii) Add $\alpha \in R$ times row j to row i: multiply on the left by $C_{ij}(\alpha) = I + \alpha E_{ij}$. Here $C_{ij}(\alpha)^{-1} = C_{ij}(-\alpha)$.

Since each elementary matrix is invertible, any sequence of elementary row and column operations transforms A into an equivalent matrix PAQ.

Definition 23 (8.7 — Matrix equivalence)**.** *Two matrices $A, B \in M_{m\times n}(R)$ are* equivalent *if there exist invertible matrices $P \in M_m(R)$ and $Q \in M_n(R)$ such that $B = PAQ$.*

Theorem 49 (8.8 — Smith Normal Form)**.** *Let $A \in M_{m\times n}(R)$ where R is a PID. Then A is equivalent to a matrix of the form*

$$\begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_r & \\ & & & 0 \end{pmatrix},$$

where $a_1, \ldots, a_r \neq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$.

Proof. Define the *length* $\ell(a)$ of a non-zero $a \in R$ to be the number of prime factors in the factorisation $a = p_1 \cdots p_k$ counted with multiplicity, with the convention $\ell(u) = 0$ for units.

Step 1: bring a minimal-length entry to position $(1,1)$. If $A = 0$ the result is immediate. Otherwise, among all non-zero entries pick a_{ij} with minimal $\ell(a_{ij})$. Row and column swaps bring it to position $(1,1)$.

Step 2: clear the first row. Fix any column $k \geq 2$. If $a_{11} \mid a_{1k}$, write $a_{1k} = a_{11}q$ and apply $C_k - qC_1$, zeroing the $(1,k)$ entry without changing a_{11}. If $a_{11} \nmid a_{1k}$, let $d = \gcd(a_{11}, a_{1k})$ and write $a_{11} = ds$, $a_{1k} = dt$. By Bézout there exist $u, v \in R$ with $us + vt = 1$, so the matrix $\left(\begin{smallmatrix} u & t \\ v & -s \end{smallmatrix}\right)$ is invertible (determinant -1). Right-multiplying A by the $n \times n$ matrix with this 2×2 block in the $(1,k)$-positions and the identity elsewhere replaces the $(1,1)$ entry by $a_{11}u + a_{1k}v = d$ and the $(1,k)$ entry by $a_{11}t - a_{1k}s = 0$. Since d properly divides a_{11}, we have $\ell(d) < \ell(a_{11})$. Repeating across all columns clears the first row.

Step 3: clear the first column. The identical argument applied to rows (left-multiplying by the analogous block matrices) clears all entries below $(1,1)$.

Step 4: termination. Clearing the first column may reintroduce non-zero entries in the first row, requiring another pass of Step 2. However this only occurs when a_{11} does not divide some first-column entry, which triggers the Bézout case and strictly reduces $\ell(a_{11})$. Since $\ell(a_{11}) \geq 0$ is a non-negative integer, the alternating process terminates after at most $\ell(a_{11})$ restarts, yielding

$$P_1 A Q_1 = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_1 & \\ 0 & & & \end{pmatrix},$$

where A_1 is an $(m-1) \times (n-1)$ submatrix.

Step 5: induction. Apply the same procedure to A_1 and continue. By induction on $m+n$ we obtain $PAQ = \operatorname{diag}(a_1, \ldots, a_r, 0, \ldots, 0)$.

Step 6: enforce $a_1 \mid a_2 \mid \cdots \mid a_r$. Suppose $a_1 \nmid a_2$. Add row 2 to row 1; the first row becomes $(a_1, a_2, 0, \ldots, 0)$. Applying Step 2 replaces a_1 by $\gcd(a_1, a_2)$, strictly reducing $\ell(a_1)$. Repeating forces $a_1 \mid a_2$, and similarly $a_1 \mid a_i$ for all i. Repeating the argument for a_2 in place of a_1, and so on, gives $a_i \mid a_{i+1}$ for all $i = 1, \ldots, r-1$. □

Definition 24 (8.9 — Invariant factors)**.** *The non-zero diagonal entries $a_1, \ldots, a_r$ of the Smith normal form of A are called the* invariant factors *of A. They are unique up to unit multipliers, and two $m \times n$ matrices are equivalent if and only if they have the same invariant factors.*

Theorem 50 (8.10)**.** *Let A be an $m \times n$ matrix over a PID. Then the row rank of A equals the column rank of A.*

Proof. Choose invertible P, Q such that PAQ is in Smith normal form (Theorem 8.8). By Theorem 8.3, multiplying by invertible matrices preserves row and column rank, so

$$\text{row rank}(A) = \text{row rank}(PAQ) = r = \text{col rank}(PAQ) = \text{col rank}(A),$$

where r is the number of non-zero diagonal entries. □

Definition 25 (8.11 — Rank)**.** *The common value of the row rank and column rank of A is called the* rank *of A, denoted* $\operatorname{rank}(A)$.

Lecture 12 — Finitely Generated Modules over a PID

Theorem 51 (9.1 — Decomposition Theorem)**.** *Let R be a principal ideal domain, and let M be a finitely generated R-module. Then*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r,$$

where $s, r \in \mathbb{N} \cup \{0\}$, the a_i are non-zero non-units in R, and $a_i \mid a_{i+1}$ for $i = 1, \ldots, r-1$.

Proof. Since M is finitely generated, Lemma 5.24 gives a surjective R-homomorphism $\phi_0 : R^n \to M$. Let $K = \ker \phi_0$. By the First Isomorphism Theorem,

$$M \cong R^n/K.$$

Because K is a submodule of the finite free module R^n and R is a PID, Lemma 5.25 says that K is free of finite rank $m \leq n$. Choose an isomorphism $\phi : R^m \xrightarrow{\sim} K$. The composite

$$R^m \xrightarrow{\phi} K \hookrightarrow R^n$$

is represented by an $n \times m$ matrix A: its jth column is the vector $\phi(e_j) \in K \subseteq R^n$, where e_j is the jth standard basis vector of R^m. Since ϕ maps onto K, the image of this matrix is exactly

$$AR^m = K.$$

By Smith Normal Form (Theorem 8.8), there are invertible matrices $P \in M_n(R)$ and $Q \in M_m(R)$ such that

$$PAQ = D = \operatorname{diag}(d_1, \ldots, d_k, 0, \ldots, 0), \qquad d_1 \mid \cdots \mid d_k,$$

where $d_1, \ldots, d_k$ are the nonzero diagonal entries. Since Q is invertible, $QR^m = R^m$: every vector in R^m is Qv for some $v \in R^m$. Therefore

$$AQR^m = AR^m = K.$$

Multiplying this equality of submodules by P gives

$$DR^m = PAQR^m = PK.$$

We now explicitly compare the quotients. Define

$$\overline{P} : R^n/K \longrightarrow R^n/PK, \qquad v + K \longmapsto Pv + PK.$$

This is well-defined: if $v + K = w + K$, then $v - w \in K$, so $P(v - w) \in PK$, and hence $Pv + PK = Pw + PK$. It is an isomorphism, with inverse $u + PK \mapsto P^{-1}u + K$. Thus, using $K = AR^m$ and $PK = PAQR^m = DR^m$,

$$M \cong R^n/K \cong R^n/PK = R^n/PAQR^m = R^n/DR^m.$$

To compute DR^m, write $v = (v_1, \ldots, v_m)^{\mathsf{T}}$. Then

$$Dv = (d_1v_1, \ldots, d_kv_k, 0, \ldots, 0)^{\mathsf{T}}.$$

The inputs $v_1, \ldots, v_k$ vary independently over R, so the first k output coordinates vary independently over the ideals $Rd_1, \ldots, Rd_k$. Hence, as a submodule of $R^n = R \oplus \cdots \oplus R$,

$$DR^m = Rd_1 \oplus \cdots \oplus Rd_k \oplus \underbrace{0 \oplus \cdots \oplus 0}_{n-k}.$$

Applying Theorem 4.33 coordinate by coordinate gives

$$\frac{R^n}{DR^m} \cong \frac{R}{Rd_1} \oplus \cdots \oplus \frac{R}{Rd_k} \oplus \underbrace{\frac{R}{0} \oplus \cdots \oplus \frac{R}{0}}_{n-k}.$$

Since $R/0 \cong R$, the last $n-k$ terms give R^{n-k}. If some d_i is a unit, then $Rd_i = R$ and $R/Rd_i = 0$, so that summand vanishes. The unit entries occur first in the divisibility chain; after deleting them and renaming the remaining nonzero non-unit entries $a_1, \ldots, a_r$, we retain $a_1 \mid \cdots \mid a_r$. Setting $s = n - k$, we obtain

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r. \qquad \square$$

Definition 26 (9.2 — Torsion element)**.** *Let R be a principal ideal domain and M an R-module. An element $x \in M$ is called a* torsion element *if there exists a non-zero $r \in R$ such that $rx = 0$.*

Definition 27 (9.4 — Torsion-free element)**.** *Let R be a principal ideal domain and M an R-module. A non-zero element $x \in M$ is called a* torsion-free element *if $rx = 0$ for $r \in R$ implies $r = 0$.*

Theorem 52 (9.6)**.** *Let R be a principal ideal domain and M an R-module. Then*

$$\operatorname{Tor} M := \{x \in M \mid x \text{ is a torsion element}\}$$

is a submodule of M.

Proof. We verify the three conditions of Definition 4.8.

Non-empty. For any non-zero $r \in R$, $r \cdot 0 = 0$, so $0 \in \operatorname{Tor} M$.

Closed under subtraction. Let $x, y \in \operatorname{Tor} M$, say $rx = 0$ and $sy = 0$ for non-zero $r, s \in R$. Then

$$rs(x - y) = s(rx) - r(sy) = 0.$$

Since R is a PID it is an integral domain, so $rs \neq 0$, hence $x - y \in \operatorname{Tor} M$.

Closed under scalar multiplication. Let $x \in \operatorname{Tor} M$ with $rx = 0$ for some non-zero $r \in R$, and let $a \in R$. Then $r(ax) = a(rx) = 0$, so $ax \in \operatorname{Tor} M$. $\square$

Theorem 53 (9.7)**.** *Let M be a finitely generated module over a principal ideal domain R. Then*

$$M = F \oplus \operatorname{Tor} M,$$

where $F \cong R^s$ for some non-negative integer s, and $\operatorname{Tor} M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$ with the a_i non-zero non-units satisfying $a_1 \mid \cdots \mid a_r$.

Proof. By Theorem 9.1, $M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$. Set $F \cong R^s$ and $T \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$, so $M = F \oplus T$. We show $T = \operatorname{Tor} M$.

$T \subseteq \operatorname{Tor} M$. We say $c \in R$ *annihilates* a module N if $cy = 0$ for every $y \in N$, i.e. multiplying by c sends everything in N to zero. We show that the single non-zero element a_r annihilates T, which makes every element of T torsion.

Step 1: $a_r \in Ra_i$ for each i. Since $a_1 \mid a_2 \mid \cdots \mid a_r$, each a_i divides a_r, i.e. $a_r = q_i a_i$ for some $q_i \in R$. Hence $a_r \in Ra_i$.

Step 2: a_r annihilates each summand R/Ra_i. An element of R/Ra_i is a coset $y + Ra_i$ with $y \in R$, and

$$a_r(y + Ra_i) = a_r y + Ra_i = y q_i a_i + Ra_i = Ra_i,$$

since $y q_i a_i \in Ra_i$ (R is commutative). The coset Ra_i is the zero element of R/Ra_i, so $a_r(y + Ra_i) = 0$.

Step 3: a_r annihilates T. An element of T is a tuple $t = (t_1, \ldots, t_r)$ with $t_i \in R/Ra_i$. Scalar multiplication acts coordinatewise, so by Step 2

$$a_r t = (a_r t_1, \ldots, a_r t_r) = (0, \ldots, 0) = 0.$$

Since $a_r \neq 0$, every $t \in T$ is a torsion element, so $T \subseteq \operatorname{Tor} M$.

$\operatorname{Tor} M \subseteq T$. Let $x \in \operatorname{Tor} M$ and write $x = x_1 + x_2$ with $x_1 \in F$ and $x_2 \in T$. Since $T \subseteq \operatorname{Tor} M$, the element x_2 is torsion, and therefore so is $x_1 = x - x_2$. Say $rx_1 = 0$ for some non-zero $r \in R$. Under the isomorphism $F \cong R^s$, x_1 corresponds to some $(y_1, \ldots, y_s) \in R^s$, and $r(y_1, \ldots, y_s) = 0$ gives $ry_i = 0$ for each i. Since R is an integral domain and $r \neq 0$, each $y_i = 0$, so $x_1 = 0$. Thus $x = x_2 \in T$. □

Theorem 54 (9.8)**.** *Let R be a principal ideal domain. Then any non-zero ideal $P \neq R$ is prime if and only if it is maximal.*

Proof. ($\Leftarrow$) Every maximal ideal is prime by Theorem 6.21.

($\Rightarrow$) Let $P = (a)$ be a non-zero prime ideal. Suppose for a contradiction that P is not maximal, so there exists an ideal M with $P \subsetneq M \subsetneq R$. Since R is a PID, $M = (b)$ for some $b \in R$. Then $(a) \subseteq (b)$ implies $a = bx$ for some $x \in R$. Since $b \notin (a)$ and (a) is prime, the relation $a = bx \in (a)$ forces $x \in (a)$, say $x = ay$ for some $y \in R$. Then $a = bx = bay$, and since R is an integral domain and $a \neq 0$, we may cancel to get $1 = by$. But then b is a unit, so $M = (b) = R$, a contradiction. □

Corollary 8. *Let R be a principal ideal domain and P a non-zero prime ideal. Then R/P is a field.*

Proof. Since P is a non-zero prime ideal in the PID R, Theorem 9.8 gives that P is maximal. It remains to show R/P is a field, i.e. that every non-zero element $a+P \in R/P$ is invertible. Since $a \notin P$, the ideal $Ra+P$ strictly contains P. As P is maximal, $Ra+P = R$, so there exist $r \in R$ and $p \in P$ with $ra+p = 1$. Then $(r+P)(a+P) = ra+P = 1+P$, so $a + P$ is invertible. □

Theorem 55 (9.9 — Uniqueness of Decomposition)**.** *Let R be a principal ideal domain and M a finitely generated R-module. Suppose*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_u \quad \text{and} \quad M \cong R^t \oplus R/Rb_1 \oplus \cdots \oplus R/Rb_v,$$

where the a_i and b_j are non-zero non-units with $a_1 \mid \cdots \mid a_u$ and $b_1 \mid \cdots \mid b_v$. Then $s = t$, $u = v$, and $Ra_i = Rb_i$ for all i.

Proof. Step 1: $s = t$. By Theorem 9.7, both decompositions yield $M = F \oplus \operatorname{Tor} M$ with $F \cong R^s$ and $F' \cong R^t$ respectively. Then

$$R^s \cong F \cong M/\operatorname{Tor} M \cong F' \cong R^t,$$

so $s = t$ by Theorem 5.21. Let $T = \operatorname{Tor} M$; it remains to show $u = v$ and $Ra_i = Rb_i$.

Step 2: $u = v$. For an R-module X and a prime $p \in R$, define $X_p := \{x \in X \mid px = 0\}$. By the corollary to Theorem 9.8, R/Rp is a field, so T_p is a vector space over R/Rp. From the first decomposition of T,

$$\left(\frac{R}{Ra_i}\right)_p = \begin{cases} R/Rp & \text{if } p \mid a_i, \\ 0 & \text{if } p \nmid a_i, \end{cases}$$

so $\dim_{R/Rp}(T_p)$ equals the number of a_i divisible by p. Now choose a prime p dividing a_1; since $a_1 \mid a_i$ for all i, p divides every a_i, giving $\dim_{R/Rp}(T_p) = u$. Applying the same reasoning to the second decomposition, the dimension also equals the number of b_j divisible by p, which is at most v, so $u \leq v$. By symmetry (choosing a prime dividing b_1), $v \leq u$, hence $u = v$.

Step 3: $Ra_u = Rb_u$. Since $a_i \mid a_u$ for all i, the element a_u annihilates every summand in the first decomposition, so $a_u T = 0$. Applied to the second decomposition, $a_u \cdot (R/Rb_u) = 0$, which means $a_u \in Rb_u$, so $Ra_u \subseteq Rb_u$. By symmetry $Rb_u \subseteq Ra_u$, hence $Ra_u = Rb_u$.

The remainder of the proof is non-examinable. It remains to show $Ra_i = Rb_i$ for $i < u$ by downward induction on i. The idea is: assuming $Ra_i = Rb_i$ for $k \leq i \leq u$, one multiplies both decompositions of T by a carefully chosen element $x \in R$ built from a prime p and the exact p-adic valuation of a_u. This zeros out the first $k-1$ summands in one decomposition but not the other, producing two decompositions of xT with a different number of non-zero summands — contradicting the fact (established by the dimension argument in Step 2) that any two such decompositions must have equally many non-zero summands. □

Theorem 56 (9.10 — Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$, the a_i are non-zero non-units in $\mathbb{Z}$, and $a_1 \mid a_2 \mid \cdots \mid a_r$. This decomposition is unique.

Proof. Every abelian group is a $\mathbb{Z}$-module (with $n \cdot a$ meaning $a + \cdots + a$ or its inverse), and $\mathbb{Z}$ is a PID. So Theorem 9.1 applies directly with $R = \mathbb{Z}$, giving a decomposition $A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z}$ where each $\mathbb{Z}/a_i\mathbb{Z} = R/Ra_i$ is a cyclic torsion summand. Uniqueness of s and of the ideals $\mathbb{Z}a_i$ (hence the a_i up to sign) then follows from Theorem 9.9.

If A is finite, then $s = 0$: the summand $\mathbb{Z}^s$ contains elements of infinite order whenever $s \geq 1$, but a finite group has no such elements. □

Lecture 13 — Rational Canonical Form

Definition 28 (The $F[x]$-module structure on V)**.** *Let F be a field, V a finite-dimensional vector space over F, and $T \in \mathrm{Hom}_F(V, V)$ a linear map with matrix A relative to some basis. We make V into an $F[x]$-module by defining, for any polynomial $f(x) = a_0 + a_1 x + \cdots + a_m x^m \in F[x]$ and any $v \in V$,*

$$f(x) \cdot v := a_0 v + a_1(Tv) + \cdots + a_m(T^m v).$$

This extends the F-action on V so that $x \cdot v = Tv$. Since any basis $\{v_1, \ldots, v_n\}$ of V over F also generates V as an $F[x]$-module, V is a finitely generated $F[x]$-module. Since $F[x]$ is a PID (as F is a field), the structure theorem of Lecture 12 applies.

Theorem 57 (10.1)**.** *Let $T \in \mathrm{Hom}_F(V, V)$, and let $f_1(x), \ldots, f_n(x)$ be the invariant factors of $A - xI$, where A is a matrix of T. Then*

$$V \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

(Proof non-examinable.) The key steps are: define the surjective $F[x]$-homomorphism $\phi : F[x]^n \to V$ by $e_i \mapsto v_i$, then show $\ker\phi = (A-xI)F[x]^n$, giving $V \cong F[x]^n/(A-xI)F[x]^n$. Since invertible changes of basis preserve this quotient up to isomorphism, one may apply Smith Normal Form (Theorem 8.8) to put $A-xI$ in diagonal form $\mathrm{diag}(f_1(x),\dots,f_n(x))$. The image of $F[x]^n$ under this diagonal matrix is $(f_1(x))\oplus\cdots\oplus(f_n(x))$, so Theorem 4.33 with $M_i = F[x]$ and $A_i = (f_i(x))$ for every i gives

$$V \cong \frac{F[x]\oplus\cdots\oplus F[x]}{(f_1(x))\oplus\cdots\oplus(f_n(x))} \cong \frac{F[x]}{(f_1(x))}\oplus\cdots\oplus\frac{F[x]}{(f_n(x))}.$$

Remark 3 (10.2)**.** *The polynomials $f_1(x),\dots,f_n(x)$ are called the* invariant factors *of A (or of T). After discarding any unit factors, the remaining non-unit invariant factors are called the invariant factors of the $F[x]$-module V.*

Definition 29 (10.3 — T-cyclic space)**.** *Let $T \in \mathrm{Hom}_F(V,V)$. If V, regarded as an $F[x]$-module via T, is cyclic, then V is called a* T-cyclic space *and T is called a* cyclic endomorphism *of V.*

Theorem 58 (10.4)**.** *Let W be a submodule of the $F[x]$-module V, and let $T \in \mathrm{Hom}_F(V,V)$ satisfy $TW \subseteq W$. Then W is a T-cyclic subspace if and only if there exists $w \in W$ such that $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W for some $k \in \mathbb{N}$.*

Proof. ($\Rightarrow$) Suppose W is T-cyclic, generated by w, so $W = F[x]\cdot w$ in the sense of Definition 4.13. Since $\dim V = n$ is finite, the vectors $w, Tw, T^2w, \dots, T^nw$ must be linearly dependent over F. Let T^rw be the first of these linearly dependent on the preceding ones, so $\{w, Tw, \dots, T^{r-1}w\}$ are linearly independent and

$$T^rw = a_0w + a_1(Tw) + \cdots + a_{r-1}(T^{r-1}w), \quad a_i \in F.$$

We claim every T^iw is a linear combination of $\{w, Tw, \dots, T^{r-1}w\}$. This is clear for $i \le r-1$ and holds for $i = r$ by the equation above. For $i > r$, apply T^{i-r} to the equation to get

$$T^iw = a_0(T^{i-r}w) + \cdots + a_{r-1}(T^{i-1}w),$$

and each term on the right is expressible in terms of $\{w,\dots,T^{r-1}w\}$ by the induction hypothesis. Since $W = F[x]\cdot w$, every element of W is of the form

$$f(x)\cdot w = (b_0 + b_1x + \cdots + b_mx^m)\cdot w = b_0w + b_1(Tw) + \cdots + b_m(T^mw),$$

using the $F[x]$-module action from the definition. Since every T^iw lies in $\mathrm{span}\{w,\dots,T^{r-1}w\}$, this set spans W. Combined with linear independence, $\{w, Tw, \dots, T^{r-1}w\}$ is a basis of W with $k = r$.

($\Leftarrow$) Suppose $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W. Any element $c_0w + c_1(Tw) + \cdots + c_{k-1}(T^{k-1}w)$ of W equals $(c_0 + c_1x + \cdots + c_{k-1}x^{k-1})\cdot w \in F[x]\cdot w$ by the module action, so $W \subseteq F[x]\cdot w$. Since $w \in W$ and W is closed under the $F[x]$-action, $F[x]\cdot w \subseteq W$. Hence $W = F[x]\cdot w = (w)$, so W is T-cyclic by Definition 10.3. □

Definition 30 (10.5 — Companion matrix)**.** *Let $f(x) = a_0 + a_1x + \cdots + a_{k-1}x^{k-1} + x^k$ be a monic polynomial over F. The* companion matrix *of $f(x)$ is the $k \times k$ matrix*

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix}.$$

Definition 31 (10.7 — Minimal polynomial of T on W)**.** *Let W be a T-cyclic subspace generated by w. Since $F[x]$ is a PID, the annihilator* $\mathrm{Ann}(w) = \{f(x) \in F[x] \mid f(x) \cdot w = 0\}$ *is a principal ideal $(f(x))$ for a unique monic generator $f(x)$. This $f(x)$ is called the* minimal polynomial *of T on W.*

Theorem 59 (10.6)**.** *Let $T \in \mathrm{Hom}_F(V, V)$ and let W be a T-cyclic subspace of V with basis $\{w, Tw, \ldots, T^{k-1}w\}$. Then the matrix of T on W is the companion matrix of the minimal polynomial $f(x)$ of T on W.*

Proof. Since $\{w, Tw, \ldots, T^{k-1}w\}$ is a basis of W, the vector $T^k w$ is a linear combination of these basis elements, say

$$T^k w = a_0 w + a_1(Tw) + \cdots + a_{k-1}(T^{k-1}w), \quad a_i \in F.$$

In terms of the $F[x]$-module action ($x \cdot v = Tv$), this says $(a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k) \cdot w = 0$, so the polynomial $g(x) := a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k$ lies in $\mathrm{Ann}(w) = (f(x))$.

Since $F[x]$ is a PID, $\mathrm{Ann}(w)$ is a principal ideal $(f(x))$ for some monic $f(x)$ of degree s. The inclusion $g(x) \in (f(x))$ gives $s \leq k$. On the other hand, $f(x) \cdot w = 0$ means $T^s w$ is a linear combination of $\{w, Tw, \ldots, T^{s-1}w\}$. Since $\{w, Tw, \ldots, T^{k-1}w\}$ are linearly independent, this forces $s \geq k$. Hence $s = k$.

Since $g(x) \in (f(x))$, write $g(x) = c \cdot f(x)$ for some $c \in F$. Both g and f are monic of degree k with leading coefficient -1 and 1 respectively, so $c = -1$ and $f(x) = -g(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_1 x - a_0$.

The matrix of T on W relative to $\{w, Tw, \ldots, T^{k-1}w\}$ has columns given by the images of each basis vector: $T(T^i w) = T^{i+1}w$ for $i < k - 1$, contributing 1s on the subdiagonal, and the last column records the coefficients $a_0, \ldots, a_{k-1}$ of $T^k w$. By Definition 10.5, this is exactly the companion matrix of $f(x)$. □

Lecture 14

Theorem 60 (10.8 — Rational Canonical Form)**.** *Let V be a finite-dimensional vector space over a field F and let $T \in \mathrm{Hom}_F(V, V)$. Then there exists a basis of V, as an $F[x]$-module, with respect to which the matrix of T is*

$$\begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_r \end{pmatrix},$$

where B_i is the companion matrix of a monic polynomial $f_i(x)$, for $i = 1, \ldots, r$, such that $f_1(x) \mid f_2(x) \mid \cdots \mid f_r(x)$. This matrix is called the rational canonical form *of T, and it is unique.*

Proof. Step 1: V is a torsion $F[x]$-module. Since $\dim_F V = n$, we have $\dim_F \mathrm{Hom}_F(V, V) = n^2$. So the $n^2 + 1$ elements $I, T, T^2, \ldots, T^{n^2}$ must be linearly dependent over F. Let T^k be the first element linearly dependent on the preceding ones, so

$$T^k = a_0 I + a_1 T + \cdots + a_{k-1}T^{k-1}, \quad a_i \in F.$$

Applying both sides to any $v \in V$ and using the $F[x]$-module action $x \cdot v = Tv$ gives

$$(a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k) \cdot v = 0.$$

Since this polynomial is nonzero, every $v \in V$ is a torsion element (Definition 9.2), so $V = \operatorname{Tor} V$.

Step 2: Decompose V using Theorem 9.1. V is a finitely generated $F[x]$-module: any F-basis of V also generates V as an $F[x]$-module, since $F[x]$ is a PID. Since $V = \operatorname{Tor} V$, the free part in Theorem 9.1 vanishes, giving

$$V = V_1 \oplus \cdots \oplus V_r,$$

where each $V_i \cong F[x]/(f_i(x))$ is a cyclic $F[x]$-module and $f_1(x) \mid f_2(x) \mid \cdots \mid f_r(x)$.

Step 3: Represent T on each V_i. Each V_i is cyclic as an $F[x]$-module, hence a T-cyclic subspace (Definition 10.3). By Theorem 10.6, there exists a basis $\mathcal{B}_i$ of V_i with respect to which $T|_{V_i}$ has matrix B_i, the companion matrix of $f_i(x)$.

Step 4: Assemble. Since $V = V_1 \oplus \cdots \oplus V_r$, the union $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$ is a basis of V, and the matrix of T with respect to $\mathcal{B}$ is $\operatorname{diag}(B_1, \ldots, B_r)$.

Uniqueness. The uniqueness of the decomposition subject to the divisibility chain, established in Theorem 9.9, implies the polynomials $f_i(x)$ — and hence the companion matrices B_i — are uniquely determined by T. □

Definition 32 (11.1 — Bi-additive map)**.** *Let M, N, and P be abelian groups. A map $f : M \times N \to P$ is called* bi-additive *if*

$$f(m_1 + m_2,\, n) = f(m_1, n) + f(m_2, n) \quad \text{and} \quad f(m,\, n_1 + n_2) = f(m, n_1) + f(m, n_2)$$

for all $m, m_1, m_2 \in M$ and $n, n_1, n_2 \in N$.

Definition 33 (11.3 — Balanced map)**.** *Let M be a right R-module and N a left R-module. A map $f : M \times N \to P$ is called* balanced *if it is bi-additive and satisfies*

$$f(mr, n) = f(m, rn)$$

for all $m \in M$, $n \in N$, and $r \in R$.

Definition 34 (11.5 — Tensor product)**.** *A* tensor product *of a right R-module M and a left R-module N is a pair (T, ϕ), where T is an abelian group and $\phi : M \times N \to T$ is a balanced map, satisfying the following universal property: for every abelian group P and every balanced map $f : M \times N \to P$, there exists a unique group homomorphism $\bar{f} : T \to P$ such that $\bar{f} \circ \phi = f$.*

Remark 4 (11.6 — Uniqueness of the tensor product)**.** *If (T', ϕ') is another tensor product of M and N, then there exists an isomorphism $\alpha : T \to T'$ such that $\phi' = \alpha \circ \phi$. Thus the tensor product is unique up to isomorphism. We write $M \otimes_R N$ for the tensor product and $m \otimes n$ for $\phi(m, n)$.*

Theorem 61 (11.2 — Existence and construction of $M \otimes_R N$)**.** *Let M be a right R-module and N a left R-module. The tensor product $M \otimes_R N$ exists. Concretely, it is realised as $T(M, N) = A(M, N)/B(M, N)$, where $A(M, N)$ is the free abelian group with basis $M \times N$, and $B(M, N)$ is the subgroup generated by all elements of the form*

$$\begin{aligned}
&(m_1 + m_2, n) - (m_1, n) - (m_2, n),\\
&(m, n_1 + n_2) - (m, n_1) - (m, n_2),\\
&(mr, n) - (m, rn),
\end{aligned}$$

for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

(Proof non-examinable.)

Lecture 15

Theorem 62 (11.7)**.** *Let M and N be right and left R-modules, respectively. Then for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$:*

(i) $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$,

(ii) $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$,

(iii) $mr \otimes n = m \otimes rn$.

(Proof non-examinable.) These follow directly from the fact that $B(M, N)$ maps to zero under ϕ, so the defining relations of $B(M, N)$ become identities in $M \otimes_R N$.

Theorem 63 (11.8)**.** *Let M and N be right and left R-modules, respectively.*

(a) For $\lambda \in \mathbb{Z}$, $m \in M$, $n \in N$: $(\lambda m) \otimes n = \lambda(m \otimes n) = m \otimes \lambda n$. In particular, $0 \otimes n = 0 = m \otimes 0$ and $(-m) \otimes n = -(m \otimes n) = m \otimes (-n)$.

(b) Every element of $M \otimes_R N$ is a finite sum $\sum_i m_i \otimes n_i$ for $m_i \in M$, $n_i \in N$.

(Proof non-examinable.) For (a), fix n; the map $m \mapsto m \otimes n$ is a group homomorphism, so it commutes with λ-multiples. For (b), every element of $A(M, N)$ is a finite $\mathbb{Z}$-linear combination $\sum \lambda_i(m_i, n_i)$, and since $\lambda_i(m_i \otimes n_i) = (\lambda_i m_i) \otimes n_i$ by (a), every element of $M \otimes_R N$ is a finite sum of simple tensors.

Remark 5 (Module structure on $M \otimes_R N$ — 11.3)**.** *If R is commutative, then $M \otimes_R N$ can be given the structure of a right R-module by defining $(m \otimes n)r = mr \otimes n$. This is well defined because R being commutative ensures the map $fr : (m, n) \mapsto mr \otimes n$ vanishes on $B(M, N)$.*

Theorem 64 (11.9)**.** *Let M be a right R-module. Then $M \otimes_R R \cong M$ as additive abelian groups. If R is commutative, then $M \otimes_R R \cong M$ as R-modules.*

(Proof non-examinable.) The map $\bar{f} : M \otimes_R R \to M$ defined by $\bar{f}(m \otimes r) = mr$ is a group isomorphism, with inverse $g : m \mapsto m \otimes 1$.

Theorem 65 (11.10)**.** *Let M, N, and P be R-modules over a commutative ring R. Then*

(i) $M \otimes_R N \cong N \otimes_R M$ *as R-modules,*

(ii) $(M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$ *as R-modules.*

(Proof non-examinable.) For (i), the map $m \otimes n \mapsto n \otimes m$ induces an isomorphism $M \otimes_R N \to N \otimes_R M$; the inverse is constructed symmetrically. Part (ii) is analogous.

Definition 35 (Tensor product of homomorphisms — §11.4)**.** *Let $f : M \to M'$ and $g : N \to N'$ be R-homomorphisms. The* tensor product $f \otimes g : M \otimes_R N \to M' \otimes_R N'$ *is the homomorphism defined by*

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

When R is commutative, $f \otimes g$ is an R-homomorphism.

Theorem 66 (11.12)**.** *Let $f : M \to M'$, $f' : M' \to M''$, $g : N \to N'$, $g' : N' \to N''$ be R-homomorphisms. Then*

$$(f' \otimes g')(f \otimes g) = f'f \otimes g'g.$$

In particular, if f and g are isomorphisms then $f \otimes g$ is also an isomorphism.

(Proof non-examinable.) Follows directly from the definition: $(f' \otimes g')(f \otimes g)(m \otimes n) = f'f(m) \otimes g'g(n)$.

Theorem 67 (11.13)**.** *Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be families of right and left R-modules, respectively. Then*

$$\left(\bigoplus_{i\in I} M_i\right) \otimes_R \left(\bigoplus_{j\in J} N_j\right) \cong \bigoplus_{(i,j)\in I\times J} (M_i \otimes_R N_j),$$

via the isomorphism $(\sum_i x_i) \otimes \left(\sum_j y_j\right) \mapsto \sum_{i,j} x_i \otimes y_j$.

(Proof non-examinable.) The map f sending $(\sum_i x_i, \sum_j y_j) \mapsto \sum_{i,j} x_i \otimes y_j$ is balanced, inducing a homomorphism $\bar{f}$ in one direction. The inclusions $\lambda_i : M_i \hookrightarrow \bigoplus M_k$ and $\mu_j : N_j \hookrightarrow \bigoplus N_\ell$ give homomorphisms $\lambda_i \otimes \mu_j$, which extend to an inverse $\bar{g}$; then $\bar{f}$ and $\bar{g}$ are mutually inverse isomorphisms.

Theorem 68 (11.14)**.** *Let R be a commutative ring, $f : M \to M'$ and $g : N \to N'$ be R-homomorphisms. If f and g are surjective, then $f \otimes g$ is surjective.*

(Proof non-examinable.) Every simple tensor $m' \otimes n' \in M' \otimes_R N'$ can be written as $f(m) \otimes g(n) = (f \otimes g)(m \otimes n)$ since f and g are surjective; hence every generator of $M' \otimes_R N'$ is in the image of $f \otimes g$.